THIS IS A SYNTHETIC WORLD: THREE ESSAYS ON CAUSAL INFERENCE, APPLIED ECONOMETRICS, AND MACHINE LEARNING FOR POLICY ANALYSIS

by

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Under the Direction of Jason Coupet, PhD

ABSTRACT

Whether we mean planning, justifying, and implementing a program for a non-profit or evaluating an already existing policy, causal inference to estimate the efficacy of policy is critical to decision making in government, business, and beyond. In general, policy analysts cannot conduct randomized trials, given their cost, ethical status, or infeasibility, and must therefore resort to carefully tuned econometric models to estimate counterfactuals, or a hypothetical scenario of how an outcome would have looked absent a given treatment or policy. Synthetic control methods (SCM) and difference-in-differences (DID) are among the most popular methods of estimating counterfactuals for policy analysts, and have seen much development in application and theory in recent years. In particular, some of their growth can be attributed to their extension via the integration of machine-learning methods. As such, my dissertation consists of three papers under a single theme: causal policy analysis in the high-dimensional setting, where *high-dimensional* refers to where we have many more control units relative to pre-treatment periods.

INDEX WORDS: Synthetic Control, Difference-in-Differences, Causal Inference, Policy Evaluation, Machine Learning

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DEDICATION

For my family, friends, and mentors.

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CHAPTER 1

Paradise Lost? Separating Policy from Pandemic in Hawaii's Border Closure with Single Proxy Synthetic Control

ABSTRACT

Hawaii's March 2020 mandatory traveler quarantine sealed the state's borders at the same instant the COVID-19 pandemic collapsed global travel demand. For a single treated unit these two forces are perfectly collinear in time, so the conventional synthetic-control toolkit, which matches the treated series to a counterfactual that is itself battered by the pandemic, cannot separate the border-closure *policy* from the coincident *pandemic*. I address this with proximal / negative-control inference in two specifications, Single Proxy Synthetic Control (SPSC) and base Proximal Inference (PI), that *instrument* the latent pandemic factor using insulated employment sectors that carry the pandemic's macro-recession shock yet are immune to the border policy. The two specifications agree, recovering a large, robust tourism-demand collapse (roughly 60–75 percentage points of year-over-year growth for the visitor-volume series); but because the insulated sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly; the estimates from insulated donors alone are therefore best read as *upper bounds in magnitude* on the policy effect and are reported as a robustness band rather than as points. Recovering an actual point requires spanning the missing travel-demand factor from outside the treated economy: adding external travel-demand donors, inbound air travel to Florida and other exposed-but-open leisure destinations that were not locked down, pulls the Visitor Days estimate off its upper bound, and a convergent ensemble of such donors

brackets it in a policy-centered band, taking the step from a bound to a band. The paper shows how negative-control inference yields credible causal estimates when a policy and a global shock arrive together.

1.1 Introduction

In 2019, tourism moved $17.75 billion in visitor spending through Hawaii's economy, underwrote $2.07 billion in state tax revenue, and sustained about 216,000 jobs, close to a third of the state's workforce (Hawai‘i Tourism Authority, 2019). In March 2020, Hawaii deliberately switched much of that off. Among all U.S. states it enacted perhaps the most extreme and comprehensive economic shutdown of the COVID-19 pandemic (Yan et al., 2022). Where most regions contracted through voluntary distancing, fear, or uneven mandates, Hawaii chose one of the few deliberate, coordinated state-level shutdowns in the country, a policy aimed not merely at slowing the virus but at suppressing travel and commerce outright. On March 26, 2020, it imposed a mandatory 14-day quarantine on all arrivals, effectively sealing its borders. The outcome was among the starkest in the country: Hawaii went on to record the largest job losses of any U.S. state even as it held among the lowest COVID-19 case rates (Bond-Smith and Fuleky, 2022). This paper asks what that decision cost.

That question is not merely historical. A governor weighing whether to seal a jurisdiction, in the next pandemic or in a climate or geopolitical shock that likewise pits containment against commerce, faces a concrete tradeoff with little credible evidence on its economic side (Newman and Noy, 2023): the health effects of COVID-19 interventions have been studied extensively, their economic toll, especially under maximal containment, far less so. Hawaii is the clearest U.S. instance of that decision in its starkest form, a whole economy switched off to keep a virus out, and a credible price for it is what a policymaker needs to weigh the closure against the lives it aimed to protect.

Estimating that cost, however, runs into an identification problem that is easy to overlook and fatal if ignored. Hawaii closed its border in the same month the pandemic collapsed global travel demand, and neither force varied before March 2020, so for a single treated unit the *policy* and the *pandemic* are perfectly collinear in time: every post-treatment month is at once a quarantine month and a pandemic month, and no function of the observed data separates a response to one from a response to the other. This disables the two designs one would reach for first. Difference-in-differences would net out the pandemic by subtracting untreated states, but that assumes the shock lands equally everywhere (de Chaisemartin and D'Haultfœuille, 2022; Roth et al., 2023), and Hawaii, the most tourism-dependent state, was struck far harder than any control. Synthetic control weakens that to a factor model and reweights donors to Hawaii's pre-2020 path, but the travel-demand collapse is dormant until March 2020 and then switches on with the policy, leaving no pre-period variation to fit: even with tourism series for all forty-nine other states, the synthetic Hawaii is a no-2020 world, not a no-quarantine-but-still-pandemic one, and differencing charges the entire post-March collapse to the border closure. Section~1.3.1 makes this formal; the failure is a property of the coincident shock, not a shortage of data.

This paper takes that identification problem seriously and resolves it with the right tool. The pandemic confounder is a time-varying latent factor that moves the outcome and coincides with treatment; it cannot be conditioned away because it is unobserved, and it cannot be matched away because it moves any donor too. The *proximal* answer (Miao et al., 2018; Tchetgen Tchetgen et al., 2024) is to instrument it: find observable negative controls that are driven by the same pandemic factor but carry no causal path from the treatment, and use them to subtract the pandemic out of the counterfactual. Here those negative controls are insulated employment sectors, hit by the pandemic recession but not downstream of whether tourists can reach Hawaii. I fit two proximal specifications, Single Proxy Synthetic Control (SPSC) (Park and Tchetgen Tchetgen, 2025) and base Proximal Inference (PI), and report both, so the estimate does not hinge on one estimator's machinery.

Empirically, the two specifications agree: fit on the insulated proxies, SPSC and PI recover the same large tourism-demand collapse, so the result rests on the identification rather than on one estimator. This delivers the first credible, if bounded, price for a maximal pandemic border closure, a collapse on the order of 60 to 75 percentage points of year-over-year growth for the visitor-volume series. What they identify is a bound, and I treat it as one. The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor, so the tourism estimates absorb both the policy and the demand loss and bound the border-closure effect from above. I report them as a specification band and check each against a relevance diagnostic. Converting that bound into a point requires a proxy that carries the missing travel-demand factor, and I supply one from outside the treated economy: inbound air travel to open, no-lockdown destinations, Florida foremost, which pulls the Visitor Days estimate off its upper bound to a policy-centered point.

The remainder of the paper proceeds as follows. Section 1.2 reviews prior research on early pandemic interventions and the methodological challenge of studying one-off, jurisdiction-wide policies. Section 1.3 states the coincident-shock identification problem and the two proximal estimators. Section 1.4 describes the data. Section 1.5 reports the proximal estimates and their identification diagnostics, the robustness band, and a set of falsification checks. Section 1.6 discusses implications for public policy and crisis response.

1.2 Background

1.2.1 Tourism in Hawaii

Hawaii's economy transitioned from a plantation-dominated model in the late 19th and early 20th centuries to tourism as the primary driver after statehood. The sugar and pineapple industries, controlled by a few powerful firms known as the "Big Five," peaked in the 1960s but declined due to global competition, labor costs, and diversification efforts. Statehood in 1959, combined with the introduction of jet air travel, spurred a tourism boom, with visitor numbers surging from 171,000 in 1958 to over 6.7 million by 1990 (Croix, 2021). This shift modernized the economy, increased population growth, and generated substantial revenue, but it also created dependency on external factors like air travel and international markets.

The scale of that dependence is worth making concrete. On any given day in 2019, Hawaii hosted about 249,000 visitors spending \$48.6 million, with O'ahu (\$22.4 million daily) and Maui (\$14.0 million) bearing the brunt (Hawai‘i Tourism Authority, 2019); leisure and hospitality employment, a proxy for tourism-related jobs, held around 190,000–200,000 pre-2020. This reliance amplifies the state's exposure to shocks even as it funds public services and infrastructure.

Hawaii implemented some of the earliest and most aggressive COVID-19 containment policies in the United States. On March 4, 2020, Governor David Ige issued an emergency proclamation officially recognizing the virus as a statewide threat (Maui Now Staff, 2020). Within weeks, the state began taking unprecedented measures alongside states like California (Friedson et al., 2021). On March 17, Governor Ige publicly urged all visitors to suspend their trips to Hawaii for 30 days, anticipating substantial economic damage to the state's tourism-dependent economy. *"We do know there will be significant impact,"* he stated, though no concrete estimates were available at the time (Nakaso, 2020). A sweeping stay-at-home order followed on March 25, banning non-essential activities, closing non-critical businesses, and effectively shutting down day-to-day life and travel across the islands (HNN Staff, 2020). These actions were implemented alongside a mandatory 14-day quarantine on all arrivals, passed the same week, which further intensified the shutdown and isolated the state from both domestic and international travel. Despite the unprecedented scope of these interventions, no formal quantitative analysis has yet estimated the short-run economic effects of Hawaii's approach. This paper provides such an assessment.

1.2.2 Literature Review

A large early literature makes the public-health case for acting fast, and the lesson is consistent: early, aggressive containment slows transmission and buys hospital capacity. Friedson et al. (2021) find that California's statewide shelter-in-place order, the first in the country, averted 160 to 195 COVID-19 cases per 100,000 and as many as 1,566 deaths within a month, at a back-of-the-envelope cost of roughly 650 to 700 jobs per life saved. Yang (2021) estimates that Wuhan's 76-day cordon cut inbound mobility by about 60% and halved the growth of new cases within days, and Cerqueti et al. (2021), using an augmented synthetic control, reach the same sooner-the-better verdict for Italy's national lockdown. Timing is what pays: Bayat et al. (2020a) estimate that locking down ten days earlier would have cut New York deaths by as much as 80%, and Sobiech Pellegrini (2022) and Li and Shankar (2023a) document the mirror image, sharp case rebounds after premature reopenings in Santa Catarina and Texas. These designs price the health payoff of containment, and the California study even prices the jobs traded for it, but they do so for interventions milder than a full border seal and lean on comparison regions that a coincident global shock would contaminate.

The economic side of the ledger is thinner, and where it exists it tends to find the damage real but transitory, and mostly not the policy's own doing. Ke and Hsiao (2022), with a panel-data counterfactual, estimate that Hubei's 76-day lockdown cut first-quarter 2020 GDP by about 37%, though the economy rebounded within months of reopening, and Gong et al. (2024) put the output cost of China's stricter 2022 zero-COVID phase at 3.9% of GDP. The recurring result is that the observed economic hit is dominated by the common pandemic shock rather than the local policy. Lee and Yang (2022) find the pandemic erased 1.1 million Korean jobs by April 2020, but only 9% of that national loss traces to regional COVID intensity, the remaining nine-tenths to a nationwide common effect; Gibson and Sun (2023) likewise attribute only 32% of the U.S. spike in new jobless claims to stay-at-home orders, the rest to the general spread of the virus and collapsing consumer confidence; and Osman et al. (2022) can identify state-level effects at all only by leaning on the cross-state policy variation that a uniform national shock erases. That decomposition is the crux of the present paper: when the observed collapse is mostly the coincident shock, a credible policy estimate must separate the two, because differencing against a control the same shock has already moved cannot. Set against these costs, containment also carried some economic and welfare upside, from the pollution-mortality co-benefits of Wuhan's lockdown (Cole et al., 2020) to the GDP losses that Europe's vaccine-certificate mandates averted (Oliu-Barton et al., 2022).

For all its breadth, this literature leaves the decision at its extreme end essentially unpriced. The question a policymaker faces is not whether containment works but what its most aggressive form costs, and that figure is what the record lacks. The gap has teeth because the politics cut against early action: as Cerqueti et al. (2021) observe, a containment that succeeds looks excessive in hindsight, so the officials who act first are precisely those most in need of a credible after-the-fact accounting to defend the call. Hawaii is the clearest U.S. instance of that most-aggressive option, and this paper supplies the missing price for a "zero-COVID'' style border (Yuan, 2022).

While much has been learned about the effects of pandemic interventions, measuring their economic impact remains hard, and Hawaii pushes that difficulty to an extreme. Most evaluations rely on difference-in-differences or synthetic control, and both lean on cross-sectional variation in policy, some counties or states acting while others do not, or acting later, to supply the untreated or not-yet-treated comparison the design needs (e.g., Yang, 2021; Gong et al., 2024; Bilgel, 2022). Callaway and Li (2023) show that even this variation is treacherous during an epidemic, because the timing-based identifying assumptions are hard to reconcile with the dynamics of contagion. And it vanishes entirely for a single treated unit hit by a global shock: every other U.S. state and comparable island economy faced its own pandemic disruption, so there is no untreated Hawaii to borrow across space. Ke and Hsiao (2023) confront a version of the same coincident-shock problem, developing a panel method for data subject to multiple treatment effects that disentangles a global pandemic's impact from a specific, coincident one, and Li et al. (2023) apply the same panel-plus-time-series logic to separate a Shanghai housing policy from the pandemic. The only prior study of Hawaii's pandemic economy, Bond-Smith and Fuleky (2022), documents the depth of the decline but compares it against a pre-pandemic forecast rather than a policy counterfactual.

One response to the missing-donor problem is to borrow strength across *time* rather than space: the Synthetic Historical Control (SHC) method of Chen et al. (2024) reconstructs the treated unit's counterfactual from its own pre-treatment trajectory. But borrowing across time does not, by itself, solve the deeper problem here. Whether the counterfactual is built from other states (classical SCM) or from Hawaii's own history (SHC), it is a world with *neither* the quarantine *nor* the pandemic; extrapolating it into 2020 and differencing against the observed series attributes the entire post-March collapse, pandemic and policy together, to the treatment. The issue is the *coincidence* of two shocks in a single unit, whatever the source of the donor information, so I do not use own-history extrapolation as the estimator and turn instead to the proximal / negative-control approach.

That is precisely the structure the *proximal / negative-control* literature was built to handle. Negative controls, variables driven by an unmeasured confounder but shielded from the treatment, have long been used to detect confounding (Lipsitch et al., 2010); proximal causal inference turns them into an identification strategy, using proxies of a latent confounder to recover causal effects that ordinary adjustment cannot (Miao et al., 2018; Cui et al., 2024; Tchetgen Tchetgen et al., 2024; Shi et al., 2020). Park and Tchetgen Tchetgen (2025) adapt this machinery to the synthetic-control setting as Single Proxy Synthetic Control, and Liu et al. (2024b) develop a related surrogate-based proximal estimator. I apply proximal inference, SPSC and base PI, to Hawaii because it is the one approach in the synthetic-control family that concedes the pandemic confounder is present and unobservable and instruments it, rather than matching or conditioning it away.

1.3 Methodology

1.3.1 The identification problem: a coincident common shock

We know how Hawaii's economy looked before the border closure, and what it did once it mandated the quarantine; but estimating the policy's cost requires a counterfactual, an estimate of how Hawaii's tourism sector would have evolved absent the closure. To build one, we adopt the comparative case study framework, comparing a single treated unit against a set of control units, or donors, that did not enact the policy. Index the N units by j, with $j = 1$ the treated unit (Hawaii) and the other N_0 the control units $\mathcal{N}_0$, and index the T months by t, collected in $\mathcal{T} = \{1, \ldots, T\}$ and split at the final pre-quarantine month T_0 into $\mathcal{T}_1 = \{t \leq T_0\}$ and $\mathcal{T}_2 = \{t > T_0\}$; y_{jt} is unit j's outcome in month t. Write y^1_{1t} and y^0_{1t} for Hawaii's month-t outcome with and without the closure. Only one is ever seen: with $d_t = \mathbf{1}\{t > T_0\}$ the treatment indicator, $y_{1t} = d_t\, y^1_{1t} + (1 - d_t)\, y^0_{1t}$.

Synthetic control builds the counterfactual as a weighted average of the donors. Stacking the treated unit's path into $\mathbf{y}_1 \in \mathbb{R}^T$ and the donors' paths columnwise into $\mathbf{Y}_0 \in \mathbb{R}^{T \times N_0}$, it returns the weight vector $\mathbf{w}$ that minimizes the pre-treatment mean squared error between the treated unit and the weighted donors,

$$\mathbf{w}^* = \underset{\mathbf{w} \in \Delta^{N_0}}{\operatorname{argmin}} \; \left\| \mathbf{y}_1 - \mathbf{Y}_0 \mathbf{w} \right\|_2^2. \tag{1.1}$$

The weights are restricted to the probability simplex $\Delta^{N_0} = \{\mathbf{w} \in \mathbb{R}^{N_0}_{\geq 0} : \sum_{j \in \mathcal{N}_0} w_j = 1\}$. Carried past T_0, the fitted values $\mathbf{Y}_0 \mathbf{w}^*$ of (1.1) are the counterfactual for the treated unit, and the quarantine's cost is the average treatment effect on the treated over the post-intervention period,

$$\text{ATT} := \frac{1}{n} \sum_{t \in \mathcal{T}_2} \left(y^1_{1t} - y^0_{1t} \right), \qquad n = |\mathcal{T}_2|. \tag{1.2}$$

Whether that estimate is credible depends on the data-generating process. The usual justification for synthetic control is the interactive fixed-effects model (Bai, 2009; Abadie et al., 2010b),

$$y^0_{1t} = \boldsymbol{\mu}_1^\top \boldsymbol{\lambda}_t + e_{1t}, \qquad y_{jt} = \boldsymbol{\mu}_j^\top \boldsymbol{\lambda}_t + e_{jt}, \quad j \in \mathcal{N}_0, \tag{1.3}$$

where $\boldsymbol{\lambda}_t \in \mathbb{R}^r$ collects time-varying factors common to all units, $\boldsymbol{\mu}_j$ collects unit-specific, time-invariant loadings, and $\mathbb{E}[e_{jt} \mid \boldsymbol{\lambda}_t] = 0$ is idiosyncratic error. Under this model, synthetic control is the standard tool, and the theory is precise about when it works. Differencing the treated series against any weighted counterfactual leaves, after some algebra,

$$y_{1t} - \sum_{j \in \mathcal{N}_0} w_j y_{jt} = \underbrace{\left(y^1_{1t} - y^0_{1t}\right)}_{\text{policy effect}} + \underbrace{\left(\boldsymbol{\mu}_1 - \textstyle\sum_{j \in \mathcal{N}_0} w_j \boldsymbol{\mu}_j\right)^\top \boldsymbol{\lambda}_t}_{\text{factor-mismatch bias}} + \left(e_{1t} - \textstyle\sum_{j \in \mathcal{N}_0} w_j e_{jt}\right), \tag{1.4}$$

the policy effect plus a factor-mismatch bias, the residual loading gap $\boldsymbol{\mu}_1 - \sum_j w_j \boldsymbol{\mu}_j$ projected onto the common factors, plus idiosyncratic noise (Abadie et al., 2010b; Ferman and Pinto, 2021; Zhang et al., 2022; Hollingsworth and Wing, 2022). The guarantee behind synthetic control is that a long pre-period with a close pre-treatment fit drives that loading gap to zero, forcing the weights to reproduce Hawaii's loadings,

$$\boldsymbol{\mu}_1 = \sum_{j \in \mathcal{N}_0} w_j \, \boldsymbol{\mu}_j. \tag{1.5}$$

For that guarantee to hold however, the same factors and loadings must govern the pre- and post-treatment periods, so that a fit achieved before treatment still holds after it. In the Hawaii case the condition fails. In March 2020 a new coordinate of $\boldsymbol{\lambda}_t$ switches on, the effect of the pandemic, absent from the data before the quarantine and present after, and so perfectly collinear with the treatment over the post-period. The weights are fit on the pre-period, where this factor is silent, so they cannot be chosen to reproduce Hawaii's loading on it: the factor-mismatch term in (1.4) no longer vanishes, and because the factor is unobserved and absent in sample, the bias cannot even be estimated. Practically, this means that even if we had noiseless, observed tourism data across the remaining United States, a good pre-treatment fit built from the donor states no longer certifies the counterfactual, and no reweighting of pre-period donors can subtract a shock the pre-period never saw.

1.3.2 Proximal inference: single-proxy and base specifications

Fortunately, one solution to this collinearity is the relatively new strand of synthetic control that employs proximal inference (Miao et al., 2018; Tchetgen Tchetgen et al., 2024; Shi et al., 2026). The idea is to find a negative control, or proxy: a variable we can use to net the pandemic out of the counterfactual and leave the policy's effect behind. For a proxy to do that, it must be both relevant and exclusionary. A relevant proxy has a strong relationship with the outcome of interest over the pre-policy period, so that it carries a genuine reading of the confounding pandemic factor. An exclusionary proxy is one the policy does not affect, so that reading stays uncontaminated by the treatment we are trying to measure. A variable with both properties moves with the pandemic but not with the border closure, which is exactly what lets the pandemic be subtracted from the counterfactual while the policy effect stays in, pulling apart two forces the collinearity had fused into one.

To state these conditions precisely, collect the candidate proxies at month t in a vector

$\mathbf{p}_t$. Relevance asks that some combination of the proxies track the treated unit's untreated outcome,

$$\operatorname{Cov}\Big(h^\star(\mathbf{p}_t),\, y_{1t}^0\Big) \neq 0 \quad \text{for some } h^\star : \mathbb{R}^{N_0} \to \mathbb{R}, \qquad t \in \mathcal{T}_1, \tag{1.6}$$

the proxy analogue of a strong first stage, whose empirical footprint is the pre-treatment fit; a proxy that does not move with y_{1t}^0 carries no reading of the factor and cannot help. Relevance and exclusion together secure a clean reading of the pandemic, but a reading is not yet a counterfactual. What connects the two is a bridge function: a function $h^\star$ of the proxies that, in expectation, equals the untreated outcome,

$$y_{1t}^0 = \mathbb{E}\Big[\, h^\star(\mathbf{p}_t) \,\Big|\, y_{1t}^0 \Big] \quad \text{almost surely}, \qquad t \in \mathcal{T}, \tag{1.7}$$

so that passing the proxies through $h^\star$ returns what Hawaii's tourism would have done absent the closure. In short, a good proxy is one the pandemic drives strongly (relevance) and the policy leaves alone (exclusion), tied to the treated counterfactual through a bridge function.

Two estimators turn the proxies into a counterfactual, differing only in how they pin down the bridge weights $\mathbf{w}$ in $h^\star(\mathbf{p}_t) = \mathbf{p}_t^\top \mathbf{w}$. Single Proxy Synthetic Control (Park and Tchetgen Tchetgen, 2025) uses the treated unit's own pre-treatment history as the instrument and fits the bridge by ridge-regularized GMM on the de-trended pre-period series,

$$\Big(\widehat{\boldsymbol{\eta}}, \widehat{\mathbf{w}}_\varrho\Big) = \underset{\boldsymbol{\eta}, \mathbf{w}}{\operatorname{argmin}} \; \Big\{ \overline{\boldsymbol{\Psi}}(\boldsymbol{\eta}, \mathbf{w})^\top \, \widehat{\boldsymbol{\Omega}} \, \overline{\boldsymbol{\Psi}}(\boldsymbol{\eta}, \mathbf{w}) + \varrho \, \|\mathbf{w}\|_2^2 \Big\}, \tag{1.8}$$

with $\overline{\mathbf{\Psi}}$ the averaged moment conditions, $\widehat{\mathbf{\Omega}}$ a weight matrix, and the ridge penalty ϱ (cross-validated) stabilizing the weights when the single instrument leaves them under-identified. Base Proximal Inference (PI) is the donors-only estimator of Shi et al. (2026), which in this implementation reduces to a least-squares fit of Hawaii's pre-period series on the donors with a GMM/HAC standard error. Either way the fitted bridge gives the counterfactual $\widehat{y}^0_{1t} = \mathbf{p}_t^\top \widehat{\mathbf{w}}$ and the ATT (1.2) as the average post-period gap $y_{1t} - \widehat{y}^0_{1t}$; the estimating equations, penalty selection, standard errors, and conformal band are given in Park and Tchetgen Tchetgen (2025) and implemented in mlsynth. Because the treated unit supplies a single instrument against the donors, the analytic standard errors are fragile here, so I lean on the specification band of Section~1.5 for uncertainty; sharing the identification but not the algorithm, the two estimators' agreement there reflects the proxy design rather than one estimator's machinery.

The proxies fall into two groups, both detailed in Section~1.4. The first is the $N_0 = 5$ insulated employment sectors: they fell in the pandemic recession, so they carry its macro-recession factor (relevance), but they do not depend on whether visitors can reach Hawaii, so the border closure leaves them untouched (exclusion). The second is a set of external travel-demand donors, inbound air travel to exposed-but-open leisure destinations (Florida foremost), which carry the pandemic's travel-demand-collapse factor while lying outside Hawaii's policy. Both enter the same proxy matrix $\mathbf{P} := \mathbf{Y}_0$, with month-$t$ row $\mathbf{p}_t$; a travel donor widens $\mathbf{P}$ to span a second factor rather than adding a control of a different kind. The split governs what the estimator can recover: the insulated sectors span only the macro-recession factor, so on them alone every tourism estimate is an upper bound in magnitude on the policy effect, absorbing the travel-demand collapse, and adding a travel-demand donor is what turns that bound into a point (Section~1.5.3). I summarize each fit by its pre-treatment root-mean-squared error $\mathrm{RMSE}_{\mathrm{pre}} = \{T_0^{-1} \sum_{t \in \mathcal{T}_1} (y_{1t} - \widehat{y}_{1t}^{0})^2\}^{1/2}$, small relative to the effect when the reconstruction is informative rather than an artifact.

1.4 Data

The analysis joins three public sources on calendar month and works throughout in year-over-year (YoY) percentage growth. The YoY transformation removes the strong seasonality of Hawaii tourism and employment, puts series measured in different units on a common scale, and lets each treatment effect be read directly as a percentage-point change in growth. Every series is stored as a raw monthly level and differenced to YoY growth at estimation time, so the panel retains levels for any later recovery analysis rather than shipping pre-differenced series. Hawaii's Department of Business, Economic Development & Tourism (DBEDT) supplies the treated tourism series and the insulated employment sectors; the U.S. Bureau of Transportation Statistics (BTS) supplies the external air-travel proxies. The estimation window is described at the end of the section.

1.4.1 Treated outcomes

The treated series are Hawaii's five tourism outcomes from DBEDT: the visitor-volume and price measures `Visitor Arrivals`, `Visitor Days`, `Occupancy`, `Mean Daily Rate`, and `Revenue per Available Room`. Each is stored as a raw monthly level (visitor counts, occupancy percentages, and dollar rates) and differenced to year-over-year growth at estimation time. Every one is downstream of the border closure: each moves when visitors' ability to arrive changes, so each can only be a treated outcome, never a control.

1.4.2 Insulated-sector proxies (the macro negative controls)

The negative-control donors are five insulated employment sectors drawn from the DBEDT seasonally-adjusted Current Employment Statistics, the monthly job count by industry (Hawai'i Department of Business, Economic Development and Tourism, 2025), expressed in YoY growth: `NatRes_Constr_Emp` (mining, logging and construction), `Wholesale_Emp` (wholesale trade), `Financial_Emp` (financial activities), `HealthCare_Emp` (health care and social assistance), and `Government_Emp`. These industries were hit by the pandemic's macro-recession shock, falling roughly 7–12% in the 2020 trough, but do not depend on whether tourists can physically enter Hawaii, so no causal path runs from the quarantine to them. They are the empirical negative controls of Section~1.3.2.[1] Figure 1.1 contrasts the two groups: the treated tourism series collapse by 50–99% at the border closure while the insulated donors dip only modestly, the visual signature of a factor structure in which the donors carry the macro shock and the tourism series carry the macro shock plus the travel-demand collapse plus the policy.

[1]The five are the most insulated supersectors in the CES taxonomy (Hawai'i Department of Business, Economic Development and Tourism, 2025). I screened the full taxonomy for the negative-control property and deliberately *excluded* the sectors that are downstream of visitor arrivals and would therefore violate it: retail trade (visitor spending), transportation, warehousing and utilities (which bundles air transportation), arts, entertainment and recreation, accommodation and food services, and real estate, rental and leasing (car and vacation rental). Empirically these fell 23–62% in the 2020 trough, tracking the tourism collapse rather than the macro recession, which is exactly the contamination that disqualifies them as controls. A second tier of genuinely insulated sectors, manufacturing, professional and business services, and information, could enlarge the donor pool, but their pre-2020 comovement with the tourism series is no stronger than the five retained donors' ($|\text{corr}| \leq 0.4$), so they do not sharpen identification of the tourism (travel-demand) factor. The one donor that does carry that factor is external, inbound air travel to Florida, added in Section 1.5.3.

1.4.3 Open-destination air travel (the travel-demand negative controls)

The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor (Section~1.3.2), so a second class of proxy is needed to carry that factor: inbound air travel to *exposed-but-open* destinations, places whose air traffic collapsed with the pandemic yet which imposed no traveler quarantine, and so register the travel-demand shock without registering Hawaii's policy. I build these from the BTS Airline On-Time Performance database (Bureau of Transportation Statistics, 2024), scraped from the agency's airport-statistics portal (https://www.transtats.bts.gov/airports.asp): monthly counts of arriving flights by airport, aggregated to a statewide total for Florida and to the individual airport for the Tampa (TPA), Orlando (MCO), and Phoenix (PHX) leisure metros, then converted to YoY growth. Florida is the archetype, a large, tourism-dependent state whose government kept its borders open through 2020, and the three metros diversify the proxy across independent leisure markets in states that likewise never locked down. Their identifying role, and why I fit them one at a time rather than en masse, is developed in Section~1.5.3.

1.4.4 Panel and estimation window

The treatment indicator `Mandatory Quarantine` switches to 1 in March 2020, giving 10 post-treatment months (March–December 2020); the post-window is capped at December 2020 so the extended raw panel cannot leak recovery months into the estimate. Although the panel ships raw levels from 1990 onward, the analysis truncates the pre-period to a recent window beginning January 2014 (74 pre-treatment months). Three decades of stale comovement is less relevant to the 2020 counterfactual than the recent history, and the 2014 window empirically sharpens the pre-treatment fit and yields non-degenerate GMM standard errors that the full-history-plus-detrend combination did not. The full-panel version is available by removing the truncation and enters the robustness band of Section~1.5.

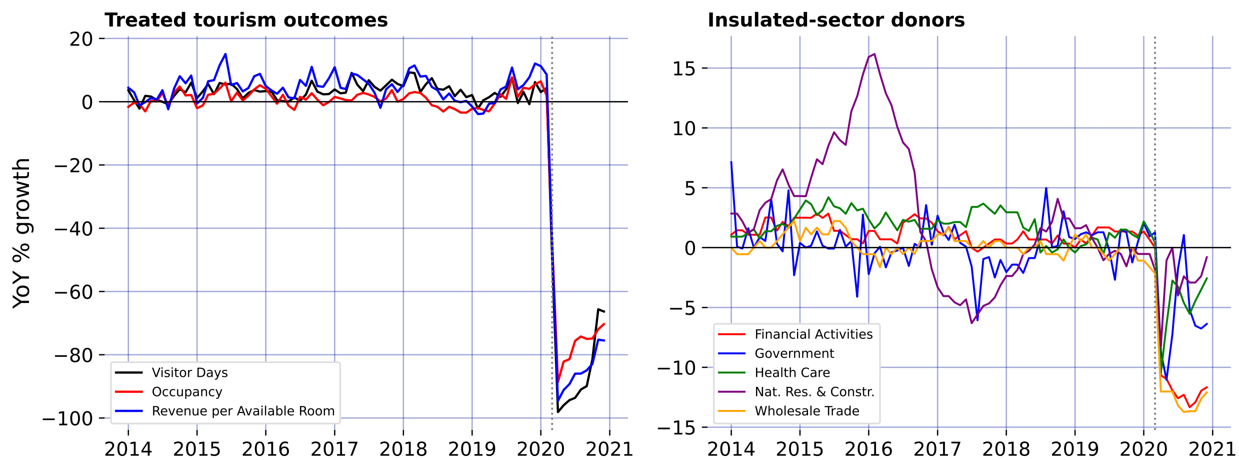


Figure 1.1: The negative-control premise: treated tourism outcomes collapse while the insulated-sector donors barely dip.

1.5 Results

1.5.1 The pre-treatment fit: relevance

Each estimate is only as good as the bridge's pre-treatment fit, the empirical footprint of the relevance condition (1.6); Figure 1.3 draws the per-outcome counterfactuals over time. Relevance holds where it matters: Visitor Days, Visitor Arrivals, Occupancy, and Revenue per Available Room all reconstruct to pre-treatment RMSEs of at most 3.6 growth-rate points (for Visitor Days, 2.6 points against a $\sim$ 70-point effect), so the collapse the estimator reads off is not an artifact of a poor fit.

The GMM standard errors are usable for the visitor series on this 2014 window ($\sim$ 3–8 points), unlike the full-history window where de-trending a near-interpolating fit collapses them toward zero. Even a well-behaved SE captures only sampling noise, not the extrapolation risk that dominates here, so I lean on the specification band below.

1.5.2 The insulated-only estimate: a bound

Fit on the insulated-sector proxies, the two headline specifications agree: for Visitor Days, SPSC returns an ATT of -68.5 percentage points and base PI -76.9, and across the five headline tourism outcomes they never differ by more than 9 points. A regularized instrumental-variable bridge (SPSC) and a plain least-squares fit (PI) landing on the same effect from the same proxies makes it a property of the proxy design, not of one estimator. But these are *upper bounds* in magnitude on the policy effect: the insulated sectors span the macro-recession factor, not the travel-demand collapse (Section~1.3.2), so the tourism loading $\boldsymbol{\mu}_1$ lies outside the donor span (1.5) along that direction and the ATT still carries it. The observed drop is policy plus travel-demand loss; closing the gap to a point needs a travel-demand donor from outside the treated economy, next.

1.5.3 Completing the bound: an ensemble of exposed-but-open destinations

The open-destination air-travel proxies documented in Section~1.4, inbound flights to Florida and to the Tampa, Orlando, and Phoenix leisure metros, all exposed-but-open, are what close the travel-demand gap that leaves the insulated-only estimates as bounds. Rather than lean on one, I use the whole small ensemble; Florida is the archetype, its air traffic down about 70% at the 2020 trough. Each enters the long panel as its own donor unit (`group = travel-demand`), and, this matters, I fit them *one at a time*, the insulated five plus a single open-destination donor, not all at once (I return to why below).

An open destination satisfies the proxy conditions from the other side: its inbound air travel loads on the same travel-demand collapse the insulated sectors miss (relevance), and as a separate economy under no Hawaii quarantine it carries that factor without importing the treatment (exclusion).

The exclusion that does the most work, though, is from Florida's *own* policy, and it is what makes Florida a deliberate choice rather than a convenient one. For the proxy to read the pandemic's voluntary travel-demand collapse cleanly, Florida's air-travel decline must not itself be an artifact of a Florida lockdown, and it is not: Florida's mandates were nominal and short-lived. Bollyky et al. (2023) measure each state's policy stringency by the share of pandemic days its mandates (business and school closures, mask and stay-at-home orders, gathering limits) were in force and place Florida at the permissive end, imposing fewer restrictions on businesses and individuals than most states (Liu et al., 2024a); the state's response was politically driven and centralized to the point of sidelining local emergency-management authority, reopening early and later barring its own localities from reimposing restrictions (Witkowski et al., 2023). Behavior matched the policy. At the March 2020 peak of the shock Florida's beaches drew spring-break crowds in the thousands even as federal guidance urged against gatherings, phone-mobility data tracked those crowds dispersing across the eastern United States (Business Insider, 2020), genomic surveillance later traced new-variant introductions to that spring-break travel (Napolitano et al., 2024), and in Pinellas County, part of the Tampa Bay leisure market that furnishes one of the donors here, ethnographers recorded beach tourism continuing through the pandemic with its protocols widely unobserved (Porter and McIlvaine-Newsad, 2022). That inbound flights to Florida collapsed anyway, with little binding policy to explain it, is precisely the point: the decline is the voluntary travel-demand factor itself, which is exactly what the proxy must carry. The criterion has teeth, and it is not met by every warm-weather destination: Puerto Rico, a fellow island otherwise tempting as a donor, imposed a genuine islandwide stay-at-home order in mid-March 2020, so its tourism collapse is contaminated by its own policy and it is set aside for that reason.

The remaining objection is that a proxy like Florida still brings idiosyncratic confounders of its own, its hurricane season, its larger drive-to share of visitors. Two features contain that objection. First, the bridge weight is fit on the clean pre-period, so a Florida-specific shock that does not track Hawaii visitor volume before 2020 enters as fit noise and is down-weighted; what licenses the weight is the pre-2020 correlation of +0.35 and the pre-treatment RMSE it earns, and the post-period behavior plays no part in the fit. Second, and decisively, the point does not rest on Florida alone: four independent markets across two states enter singly and converge on the same band, so for an idiosyncratic confounder to move the estimate it would have to be common to all four and aligned with the March 2020 quarantine, which is the pandemic travel-demand collapse itself, the very factor the donors are meant to carry. Where a proxy fails relevance the procedure reports it rather than forcing a fit, as with Occupancy below, where Florida is down-weighted to zero.

Figure 1.2 reports the convergence, and the headline is robustness through agreement. For Visitor Days, every one of the four independent open-destination donors pulls the estimate off the insulated-only bound of -68.5 points into a policy-centered band of roughly -52 to -65 points, and each one *improves* the pre-treatment fit rather than degrading it. That four proxies chosen independently, different cities, two different states, move the estimate the same way and by a similar amount is far stronger evidence that the observed drop is policy plus a real, separable travel-demand collapse than any single proxy could be. Florida gives the cleanest single fit (its Visitor Days pre-treatment RMSE falls from 2.58 to 1.92, bridge weight $w_{\mathrm{FL}} = 0.14$), consistent with its pre-2020 correlation of +0.35 with Hawaii visitor volume.

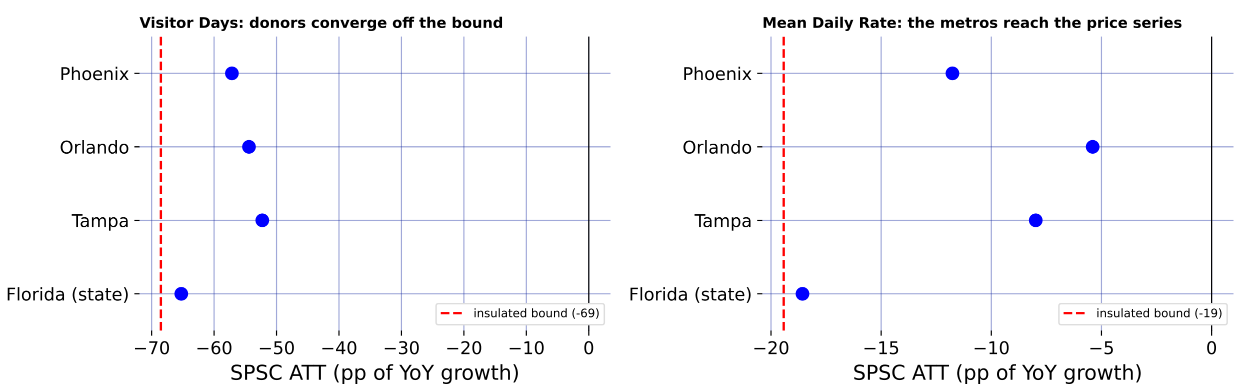


Figure 1.2: The open-destination ensemble: each donor, fit singly, pulls the estimate off the insulated-only bound.

The ensemble also reaches an outcome a single volume proxy could not. Florida is a flight-*volume* series, so it comoves with Hawaii *visitor volume* but not with hotel prices, and leaves Mean Daily Rate essentially at its insulated-only value of -19.4 points. Phoenix, a desert-resort market whose room rates swing with leisure demand, does comove with Hawaii's daily room rate (pre-2020 correlation +0.36), and adding it pulls Mean Daily Rate from -19.4 toward -11.8 points while *improving* the fit (pre-treatment RMSE 2.38 to 1.82). Different open destinations carry the travel-demand factor as it manifests in different outcomes, volume for the Florida metros, price for Phoenix, so a small, well-chosen ensemble reaches further into the tourism block than any one proxy.

This is why I fit the donors *singly*. Added en masse, the four near-collinear proxies induce large offsetting ridge weights that *degrade* the fit where a single well-chosen donor helped most, the price series (Revenue per Available Room's pre-treatment RMSE rises from 3.61 to 4.41). Independent proxies read one at a time corroborate as a convergence band; collapsed into one super-donor, the collinearity punishes them.

Figure 1.3 is the summary figure: for every headline outcome it draws the observed series, the insulated-only SPSC counterfactual (macro factor only, the upper bound, with its conformal spikes), the counterfactual after adding Florida (travel-demand factor as well), and base PI's insulated path as a dotted overlay sitting almost on the SPSC insulated line, making the two-specification agreement visible. The wedge between the two SPSC lines is where the travel-demand donor works, wide for the visitor-volume series, absent for Occupancy where Florida is correctly down-weighted to zero. No single wedge is a clean decomposition, though; each +donor fit re-solves all weights jointly, so the honest object is the convergence across donors in Figure 1.2.

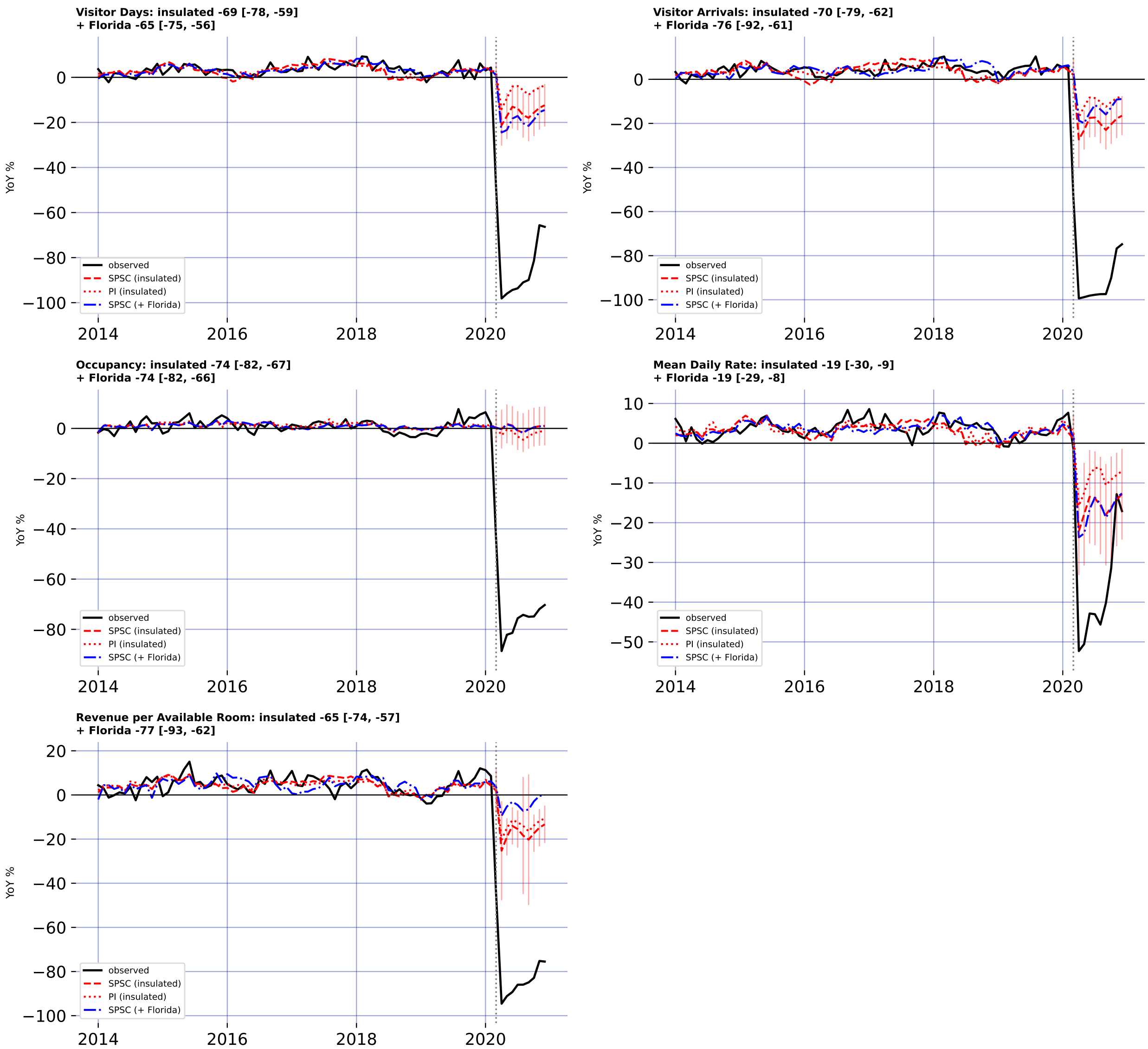


Figure 1.3: Observed, SPSC and PI insulated counterfactuals, and SPSC + Florida, for five outcomes.

1.5.4 Robustness and falsification

1.5.4.1 A specification band, not a point

Because SPSC and base PI leave the bridge weights unconstrained, dropping the simplex restriction for the flexible $\mathbb{R}^{N_0}$ of Doudchenko and Imbens (2016), the counterfactual can extrapolate outside the donor hull, and the GMM sandwich, degenerate with a single instrument against five donors on a near-interpolating pre-window, reports a sampling noise that collapses toward zero. The uncertainty that actually binds is this extrapolation, or specification, risk, so I report the headline effects as a band built from specification sensitivity: de-trending on versus off, the B-spline degrees of freedom, the polynomial basis degree, dropping each of three insulated donors individually (the two largest-weight sectors and construction, the proxy most plausibly linked to tourism), and the full-1991 versus 2014 window. Figure 1.4 collects the minimum, median, and maximum SPSC ATT across these eight specifications for the three best-behaved outcomes.

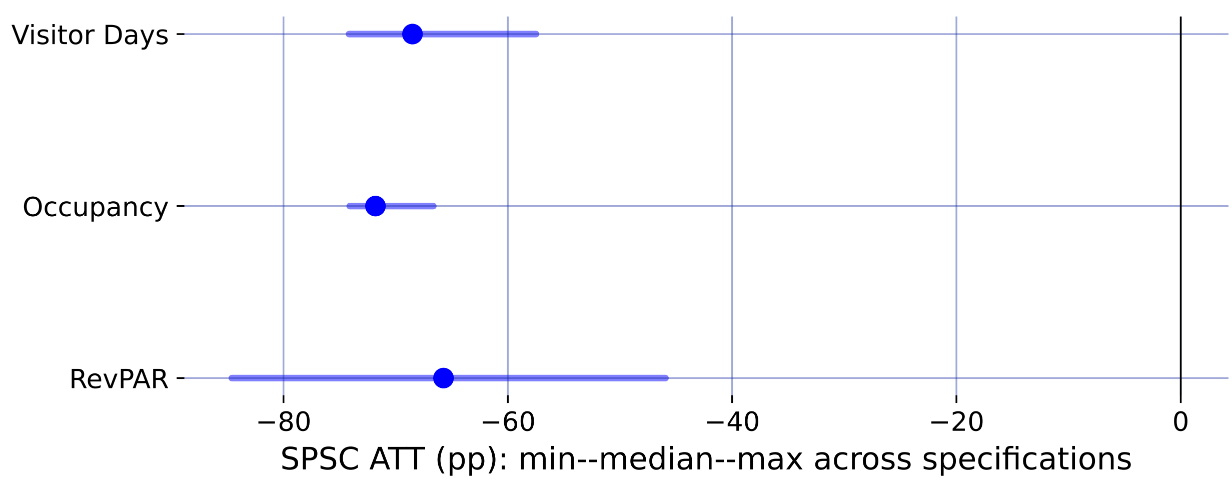


Figure 1.4: Specification robustness band: median SPSC ATT (point) and min–max range across eight specifications.

For the visitor-volume series the band runs roughly from -74 to -57 points (median -68), so the defensible insulated-only statement is a $\sim$ 60–75-point upper-bound collapse rather than any single figure.

1.5.4.2 How much would an exclusion violation have to bite?

The identifying content of the insulated proxies rests on exclusion: the border policy must not move them directly (Section~1.3.2). That is an assumption. Rather than take it on faith I bound the consequence of its failure, a sensitivity analysis in the spirit of proximal partial identification (Ghassami et al., 2023) that here follows the plausibly-exogenous relaxation of an exclusion restriction (Conley et al., 2012): posit a direct policy effect on the proxies and carry it through to the estimand. Suppose it fails: general-equilibrium spillovers from the closure depress or inflate the insulated sectors, so that over the post-period $p_{jt} = \boldsymbol{\mu}_j^\top \boldsymbol{\lambda}_t + \delta_j d_t + e_{jt}$, with δ_j the direct policy effect on proxy j and $\boldsymbol{\delta} = (\delta_j)_{j \in \mathcal{N}_0}$. Because the bridge weights are estimated over the pre-period, where $d_t = 0$ and the contamination is absent, $\widehat{\mathbf{w}}$ is untouched and the violation reaches the estimate only through the post-period counterfactual, as the exact bias

$$\widehat{\text{ATT}} = \text{ATT} - \boldsymbol{\delta}^\top \widehat{\mathbf{w}}, \tag{1.9}$$

so the bias is linear in the contamination and known up to $\boldsymbol{\delta}$. This makes the sensitivity fully transparent.

For Visitor Days the fitted weights sum to 1.92, so a common spillover of δ points moves the ATT by $1.92\,\delta$. Driving the insulated-only estimate of -69 points to zero would take $\delta \approx$

+36 percentage points, the wrong sign (a border closure that *raised* health-care, finance, and government employment growth by nearly forty points) and larger than any insulated sector's *entire* observed 2020 movement, which reached at most 12 points and is itself mostly the macro recession the proxies are meant to carry, not policy spillover. In the economically natural direction, the policy depressing the insulated sectors, $\delta_j < 0$, the bias runs the other way: charging each sector's *full* 2020 decline to the border policy moves the estimate to -83 points, so the reported effect would if anything *understate* the truth. Even an adversarial assignment that signs every δ_j to mask the effect, bounded by each sector's observed movement, shifts the ATT by at most 18 points. Occupancy is more robust still: its weights sum to essentially zero (+0.03), so a common exclusion violation cancels outright. The conclusion does not require the insulated sectors to be perfectly insulated, it survives any exclusion violation of plausible magnitude, and the one plausible direction makes it conservative.

Equation (1.9) treats the insulated sectors symmetrically, but one is more suspect than the rest: construction tracks hotel building, so it is the insulated proxy most plausibly linked to tourism and the one a reader is likeliest to distrust. Dropping it outright is the direct negative-control check. Construction is not idle in the bridge, carrying a weight of -0.35 on Visitor Days, yet removing it leaves the estimate at -73 points against -69 with all five sectors, and the pre-treatment fit is if anything slightly better (RMSE 2.52 versus 2.58), because the four remaining sectors span the same macro-recession factor and absorb its removal. The direction is the telling part: the estimate moves *away* from zero, not toward it. Were construction contaminated by the tourism collapse, its co-movement with the treated series would pull the counterfactual down and bias the effect toward zero, so purging it would enlarge the effect, which is exactly what happens. The headline does not lean on the one donor a skeptic would challenge first.

1.5.4.3 A placebo in time

A distinct worry concerns the estimator itself: does the ridge-GMM bridge, with its spline de-trending, manufacture a large effect out of ordinary pre-period variation? The standard falsification is an in-time placebo, assign a fake border closure in a pre-pandemic March, truncate the sample before 2020 so no real shock is in view, and refit. A design that recovers effects only where they exist should return roughly zero.

It does. Across Visitor Days, Occupancy, and Revenue per Available Room, and fake treatment dates in March of 2017, 2018, and 2019, the placebo ATTs never exceed 4 points in magnitude, the scale of the pre-treatment RMSE, and an order of magnitude below the $\sim$ 70-point effects the same procedure returns for 2020. The bridge does not fabricate a collapse out of business-as-usual fluctuation, so the 2020 estimate is a property of the treatment, not of the estimator.

1.5.5 Who bore the cost: incidence across islands and hotel tiers

The DBEDT workbook that supplies the statewide hotel outcomes also reports them by county and, for revenue per available room, by hotel class, so the same proximal machinery can ask not only how large the tourism collapse was but who absorbed it. For each disaggregated series I refit SPSC on the identical insulated donors and, as in Section~1.5.3, on the insulated set plus Florida; Figure 1.5 collects the estimates. Three readings stand out.

First, the demand shock was close to spatially uniform. Hotel occupancy fell between 65 and 76 points across all four counties, so the intuition that the tourism-dependent neighbor islands suffered a deeper demand hit than Oahu is not borne out in occupancy: the rooms emptied everywhere at once.

Second, the collapse ran through volume, not price. The average daily room rate fell only 12 to 24 points, a fraction of the occupancy drop, so hotels largely held their posted rates while rooms went unsold. Because revenue per available room is occupancy times rate, the revenue loss is almost entirely the occupancy loss.

Third, the revenue burden fell regressively across the market. RevPAR by hotel class traces a clear gradient: the luxury and upper-upscale tiers lost the most (90 and 99 points), the upper-midscale tier markedly less among the well-fit segments (56), with midscale and economy least affected of all. Discretionary luxury-leisure travel evaporated where lower-tier and essential lodging demand partly persisted, so the establishments and workers tied to the top of the market bore the sharpest revenue hit.

These disaggregated panels are noisier than the statewide aggregates, and several cells (faded in Figure 1.5, pre-treatment RMSE above six growth-rate points) carry the near-zero GMM standard errors that Section~1.3.2 flags as fragile when the proxies supply one instrument against the donors. I read those cells as directional only; the three patterns above rest on the well-fit segments and hold under both the insulated bound and the Florida-adjusted point.

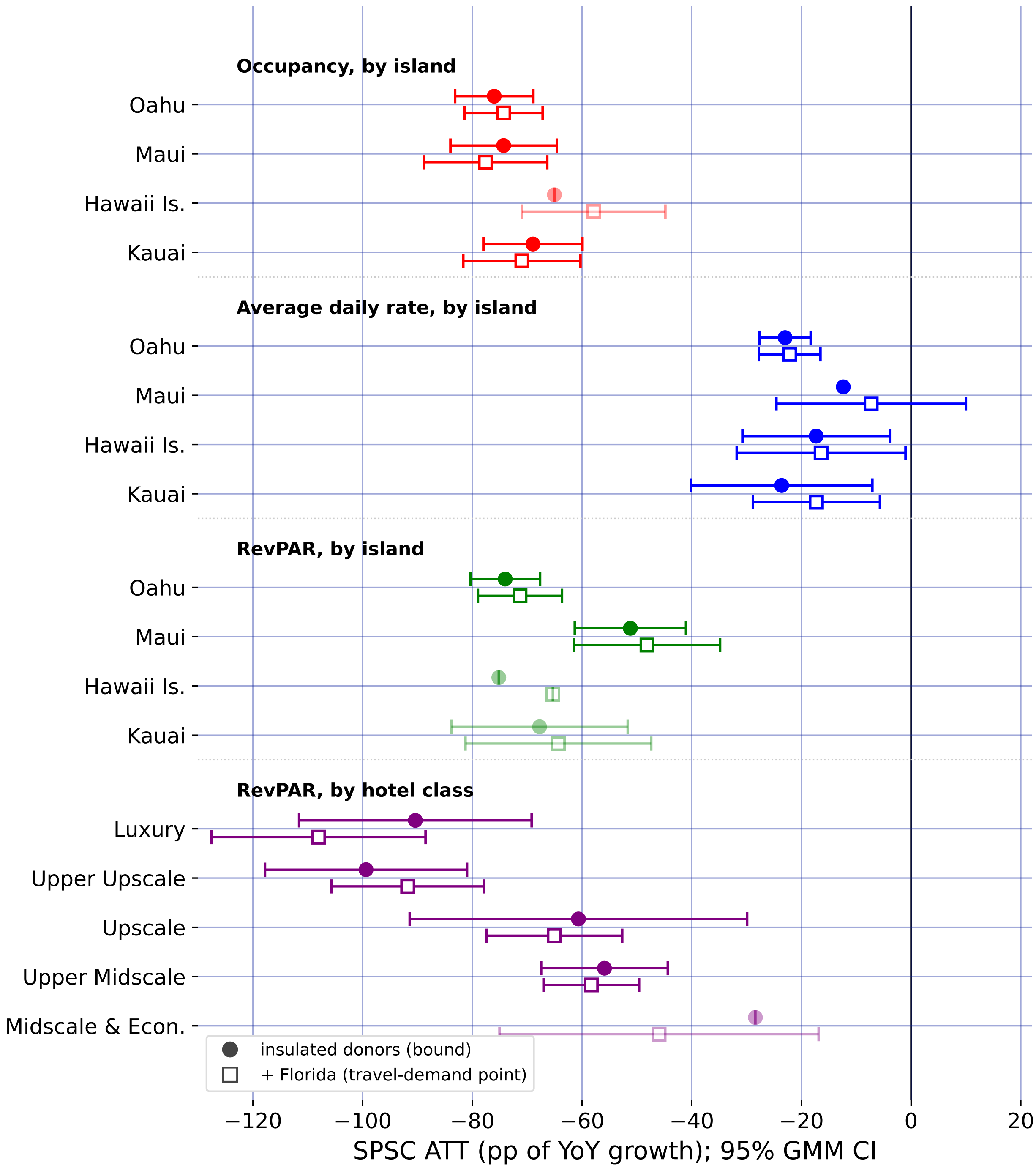


Figure 1.5: Subgroup ATTs by island and hotel class: insulated bound (circles), Florida-adjusted point (squares).

1.6 Discussion

The methodological lesson of Hawaii is that a good pre-treatment fit is not identification when a policy and a global shock arrive in the same month. A synthetic control that matches Hawaii to any pre-pandemic counterfactual, whether by borrowing across states or, as own-history extrapolation does, across the treated unit's own past, produces a clean-looking counterfactual and a large ATT, but that ATT is the border-closure effect plus the pandemic's contribution to Hawaii's outcomes, and nothing in the fit reveals the mix. Proximal / negative-control inference is what separates them: instrumenting the latent pandemic factor with insulated employment sectors that carry it but are immune to the policy, both headline specifications, SPSC and base PI, net out the macro-recession component and agree on the result, so the estimate rests on the identification rather than on any single estimator.

At the same time, the analysis is candid about the limit of what the shipped data can identify. The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor: even with an open border, Hawaii tourism would have fallen sharply as people stopped flying. SPSC on insulated sectors alone therefore recovers only the macro component of the counterfactual, and the resulting tourism ATTs over-attribute the collapse to the policy; they are upper bounds in magnitude, which is why I report them as a band rather than as points. Recovering an actual point estimate from insulated donors alone is not possible here, because every treated series loads on the travel-demand factor those donors miss; the point has to be spanned from outside the treated economy, which is what the open-destination donors do. The substantive conclusion is thus deliberately modest and, I would argue, more credible for it: the border closure coincided with a near-total collapse in measured tourism demand, of which a bounded share is causally the policy, with a convergent ensemble of external travel-demand donors pulling the Visitor Days estimate off its upper bound toward a policy-centered band.

Section~1.5.3 takes the decisive step of closing the two-factor gap with external travel-demand negative controls (inbound air travel to Florida and to a set of other exposed-but-open leisure markets that did *not* lock down), and the result is instructive about both the promise and the care the approach demands. For Visitor Days the open-destination donors do what the design promises, and they do it *in agreement*: fit one at a time, all four independently pull the estimate off the insulated-only bound into a policy-centered band, each improving the pre-treatment fit. Because the donors are chosen independently (different cities, two different states), their convergence is far stronger evidence that the missing factor is real and separable than any single proxy could be. A well-chosen ensemble also reaches further than one volume series: Phoenix, whose resort room rates track leisure demand, pulls the daily-room-rate series that Florida (a volume proxy) leaves essentially untouched. But the ensemble does not mechanically dissolve every bound, and the search must stay disciplined: near-collinear open destinations added *en masse* induce large offsetting weights and hurt the fit, so the donors are best read as a corroborating convergence band; and Puerto Rico, tempting as a fellow island, fails as a control because it imposed its own stay-at-home order in mid-March 2020 and is a contaminated, treated-like unit that, if anything, bounds the effect from the other side.

These estimates are one side of a ledger. They price the short-run economic cost of the border closure through the end of 2020; they do not, on their own, value the public-health benefit (lives saved, hospital strain averted) the closure was meant to buy. A bounded cost, though, is exactly the input a cost-benefit calculation needs, and the surrounding literature supplies the other side. Work on early containment finds that acting even days sooner averts a large share of deaths: Bayat et al. (2020a) estimate that locking down ten days earlier could have cut New York deaths by more than 80%. Evidence on comparably aggressive closures finds the economic damage sharp but short-lived (Ke and Hsiao, 2022). Set side by side, the calculus for a decisive closure is a large but transitory economic loss weighed against a benefit measured in lives, and this paper supplies the first credible, bounded estimate of that loss for the U.S. case. The incidence analysis of Section~1.5.5 adds that the loss fell unevenly: hardest on the upper tiers of the lodging market, and through lost volume rather than lower prices. What the paper also provides is a template for pricing such policies at all: when the intervention and the crisis it responds to are one and the same event in time, no comparison unit is clean, and the analyst's task is to find a variable that sees the crisis but not the policy.

1.7 Conclusion

This paper estimates the short-run economic consequences of Hawaii's March 2020 mandatory quarantine while confronting the identification problem that its predecessors on this topic could not: for a single treated unit, the border-closure policy and the COVID-19 pandemic are perfectly collinear in time, so no synthetic control that merely matches Hawaii to a pre-pandemic counterfactual can separate the two. I resolve this with proximal inference (negative-control estimators) in two specifications (Single Proxy Synthetic Control and base Proximal Inference) that instrument the latent pandemic factor using insulated employment sectors which carry the pandemic's macro-recession shock but are immune to the border policy.

Estimated by both specifications, the insulated-sector proxies identify a large, robust tourism-demand collapse (about 60–75 points of year-over-year growth for the visitor-volume series). But because those sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly: the insulated-only estimates are upper bounds in magnitude on the policy effect, reported as a robustness band rather than as points. Because every treated series loads on the travel-demand factor those donors miss, the trustworthy point comes only once the missing factor is spanned from outside the treated economy, the role of the external open-destination travel-demand donors.

Methodologically, the study demonstrates how negative-control inference yields credible causal estimates when a policy and a global shock coincide, a structure common to pandemic border closures, but also to climate shocks and other systemic crises where a would-be counterfactual unit is itself engulfed by the event. Substantively, it delivers a bounded, honest estimate of the economic price of Hawaii's "Great Isolation with Aloha characteristics'". Adding a small ensemble of external no-lockdown travel-demand donors (inbound air travel to Florida and to the Tampa, Orlando, and Phoenix leisure markets) takes the step from a bound to a policy-centered band: fit independently, they converge in pulling the Visitor Days effect toward the policy-only magnitude, and one of them (Phoenix) reaches the room-rate series a single volume proxy cannot. Finishing the job for every outcome asks only for more such open-destination proxies, disciplined by relevance, which the design readily accommodates.

CHAPTER 2

Clearing the Air: How India's 2020 Lockdown Impacted Air Quality

2.1 Introduction

The COVID-19 pandemic prompted governments worldwide to implement non-pharmaceutical interventions (NPIs), such as mass lockdowns, to curb virus spread and protect public health systems. India's nationwide lockdown, beginning on March 25, 2020, was among the strictest globally since the Wuhan lockdown in China, halting mobility, industry, and construction for 1.3 billion people. Early research focused on virus outcomes (Yang, 2021; Bayat et al., 2020a) and economic impacts (Gong et al., 2024; Wu et al., 2023), but lockdowns also likely influenced air quality, a critical public health factor. Popular media reported improved air quality during lockdowns, later confirmed by studies documenting reductions in pollutants such as PM2.5 (Watts and Kommenda, 2020; Venter et al., 2020; Kumari and Toshniwal, 2021). Given that India has some of the most polluted cities in the world, understanding how large-scale interventions affect pollution levels is of great interest. However, many studies are descriptive and do not estimate counterfactual scenarios—what air quality would have been without lockdowns—limiting their policy relevance, or rely on strict assumptions for identification (Milman and Uteuova, 2025).

This study quantifies the causal impact of India's 2020 national lockdown on PM2.5 levels nationally and in major cities using the Synthetic Historical Control (SHC) method (Chen et al., 2024). Unlike studies that rely on untreated comparison groups—which are unviable in the COVID-19 context—or on covariate conditioning, I develop an extension of SHC that accommodates imperfect pre-treatment fit. By constructing counterfactuals from historical data with flexible, non-parametric methods, SHC isolates the lockdown's effects without requiring unexposed control units. In addition to the national analysis, this study examines Delhi, Mumbai, Bangalore, and Kolkata—megacities consistently among the world's most polluted, and major hubs of finance and commerce. Embedding these cases in a unified causal framework moves beyond simple pre–post comparisons and addresses a key policy question: what happens to air pollution when a country of India's scale implements a nationwide shutdown? These findings can inform sustainable air quality management in India and provide lessons for urban planning and climate change mitigation globally.

The rest of the paper is organized as follows. Section 2.2.1 provides background on air pollution in India, highlighting its health and economic costs. Section 2.2.2 summarizes existing literature and methodological gaps. Section 2.2.2.2 outlines the empirical strategy, including SHC and the augmented version proposed here. Section 2.3 details data collection and processing.

2.2 Background and Previous Work

2.2.1 Policy Context

Air pollution is one of India's most pressing environmental and public health challenges. This section provides context on pollutant sources, exposure levels, health and economic impacts, and how the COVID-19 lockdown created a natural experiment for studying air quality.

2.2.1.1 Sources and Severity of Air Pollution

Fine particulate matter (PM2.5), which penetrates deep into the lungs and bloodstream, is a primary pollutant of concern. Major contributors include vehicular emissions, industrial activities, construction dust, biomass burning for cooking and heating, and agricultural residue burning, particularly in northern states during winter. Natural factors such as dust storms and seasonal weather patterns exacerbate pollution levels. Rapid urbanization and population growth have intensified exposure, affecting over 1.4 billion people.

Recent data highlight the severity: in 2024, India's average annual PM2.5 concentration was 50.6 $\mu g/m^3$, far exceeding the WHO guideline of 5 $\mu g/m^3$ (Statista, 2025). Regional variation is significant—cities in the Indo-Gangetic Plain, like Delhi, often exceed 100 $\mu g/m^3$ during peak winter, while southern cities such as Chennai frequently surpass both WHO and national standards (Bhuyan et al., 2025; Singh et al., 2021).

2.2.1.2 Health Impacts

The health consequences of chronic PM2.5 exposure are substantial. Long-term exposure increases the risk of respiratory and cardiovascular diseases, including COPD, asthma, lung cancer, stroke, and heart disease (Joshi et al., 2025). In India, air pollution causes roughly 1.5 million premature deaths annually, with each 10 $\mu g/m^3$ rise in PM2.5 linked to an 8.6% increase in mortality (Jaganathan et al., 2024). Vulnerable populations, namely children, the elderly, and those with weaker immune systems, are most affected. Fine particulate pollution reduces average life expectancy by 5.3 years compared to WHO guidelines, with residents in

Delhi facing up to a 10-year reduction (BBC, 2022). Short-term PM2.5 spikes also increase daily death rates by 1.42–3.57% per 10 $\mu g/m^3$ (de Bont et al., 2024).

2.2.1.3 Economic Impacts

Air pollution imposes significant economic costs through healthcare expenditures, lost labor productivity, and impacts on sectors such as tourism and agriculture. Estimates suggest annual losses of USD 95 billion, roughly 1.36% of GDP (India State-Level Disease Burden Initiative Air Pollution Collaborators, 2020; Gupta and Damani, 2021). In the Delhi NCR, productivity declines and reduced tourist arrivals contribute substantially to these costs (Dalberg Advisors, 2021; Our Common Air Commission, 2024). Efforts like the National Clean Air Programme (NCAP), launched in 2019, aim to reduce PM2.5 and PM10 concentrations by 20–30% in 131 non-attainment cities by 2024–25 (Kawano et al., 2025).

2.2.1.4 The COVID-19 Lockdown as a Natural Experiment

The 2020 nationwide COVID-19 lockdown provides a unique opportunity to study rapid air quality improvements. Imposed on March 25, the lockdown halted transportation, industrial operations, and construction, leading to reductions in PM2.5 by 31–43%, with some cities experiencing drops up to 50% (Saleem et al., 2024; Nigam et al., 2021; Mishra et al., 2021). In megacities like Delhi and Mumbai, the Air Quality Index (AQI) improved by as much as 58%, primarily due to lower vehicular and industrial emissions (Zhang et al., 2020).

Although temporary, these reductions highlight the potential of targeted interventions. PM2.5 reductions were most pronounced in early lockdown phases, averaging 30–50% across monitored sites, with occasional spikes due to meteorology (Hu et al., 2022; Goel et al., 2021; Wang et al., 2020). Other pollutants, including NO2 and PM10, also decreased, consistent with global lockdown trends (Venter et al., 2020). Understanding these dynamics is crucial for causal evaluation and future policy planning.

2.2.1.5 Lockdown Policy Phases

India's lockdown, enacted under the Disaster Management Act of 2005, unfolded in multiple phases. Phase 1 (March 25–April 14) involved a complete nationwide shutdown, prohibiting non-essential movement, closing factories, and suspending public transport. Phase 2 (April 15 onwards) allowed a gradual reopening of essential services and agriculture, with color-coded zoning based on infection rates (Deshpande, 2022; Ranjan et al., 2020). While primarily a public health measure, the lockdown served as a large-scale quasi-experiment in emission control, revealing the disproportionate contribution of human activities to PM2.5 levels.

2.2.2 Literature Review

2.2.2.1 Methodological Aspects of Prior Literature

To contextualize our approach, we review existing studies on lockdown-related air quality changes and highlight key methodological gaps. Early studies primarily relied on descriptive pre–post comparisons or satellite imagery without causal adjustment (Mahato et al., 2020; Pathakoti et al., 2021; Srivastava et al., 2020; He et al., 2020; Wang et al., 2020). These approaches document reductions in pollutants but cannot answer counterfactual questions—what would air quality have been absent the lockdown? Some studies have applied causal models. For example, Heffernan et al. (2025) use causal machine learning and interrupted time series (ITS) methods to find reductions in NO2 levels. ITS was applied in Italy (Cameletti, 2020), finding no meaningful changes in air pollutants. Augmented Synthetic Control methods were used for Wuhan (Cole et al., 2020), showing a 63% reduction in NO2. Despite these examples, a recent systematic review finds that most lockdown–air pollution studies remain descriptive or non-causal (Silva et al., 2022), limiting their policy relevance.

2.2.2.2 Gaps and Contribution of This Study

A central gap is the lack of causal designs for national-scale interventions. Many studies use simple pre–post t-tests (Goel et al., 2021), graphical analyses (Kumari and Toshniwal, 2020; Benchrif et al., 2021), year-over-year comparisons (Le Quéré et al., 2020), or OLS regressions (Pal et al., 2022). These methods cannot fully account for confounding trends or temporal patterns, reducing confidence in policy-relevant estimates. SCM improves on this by constructing a counterfactual from untreated donor units as a weighted average (Ben-Michael et al., 2021a). However, SCM is limited when studying national policies, where no valid donor pool exists, given that SCM requires untreated control units in order to be viable. Similarly, causal machine learning approaches (Heffernan et al., 2025) rely on strong ignorability assumptions that may not hold for air pollution data with unmeasured year-specific factors. In contrast, the SHC method avoids these limitations by constructing counterfactuals directly from historical patterns, without assuming complete covariate measurement. This study extends SHC to an Augmented SHC (ASHC) framework, allowing limited extrapolation to improve pre-treatment fit while maintaining stability. By applying ASHC to India's national and city-level PM2.5 data, this work provides the first rigorous causal estimates of the 2020 lockdown's air quality impacts in a country of India's scale.

Methodology

Below I describe the Synthetic Historical Control (SHC) methodology from Chen et al. (2024). I then extend this framework to settings where pre-treatment fit is poor, yielding the Augmented Synthetic Historical Control (ASHC). The ASHCM is applied separately to the national average of PM2.5 for India, as well as Delhi, Mumbai, Bangalore, and Kolkata.

2.2.3 The Synthetic Historical Control Method

SHC estimates the causal effect of an intervention by constructing counterfactual outcomes from historical segments of the treated unit itself, rather than from contemporaneous control units. This makes it well-suited for national-scale interventions such as India's 2020 lockdown, where no valid donor pool of untreated countries exists. Before presenting the estimator, I state the assumptions that guarantee SHC's validity.

2.2.3.1 Assumptions

Assumption 1 (Error Properties)**.**

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t^2] < \infty \qquad \forall\, t \in \mathcal{T}. \tag{2.1}$$

This ensures that noise is well-behaved. The mean-zero condition rules out systematic bias, while the finite variance condition prevents extreme fluctuations.

Assumption 2 (Latent Smoothness and Matching)**.**

$$\widetilde{y}(t) \in C^{H+1}, \quad H \geq 3, \qquad \sup_t \left| \frac{d^k}{dt^k} \widetilde{y}(t) \right| < \infty \ \ \forall\, k \leq H+1, \tag{2.2}$$

and there exists

$$\mathbf{w}^* \in \Delta^{N-1} \qquad \textit{such that} \qquad \mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}^*. \tag{2.3}$$

Smoothness ensures that the latent process can be approximated by a finite expansion. The matching condition guarantees that some convex combination of historical donor segments can reproduce the evaluation period. This parallels the "existence of weights" assumption in the SCM literature (Abadie et al., 2010b; Amjad et al., 2018). Importantly, persistent factors such as meteorology are implicitly matched so long as they appear in the historical trajectory.

Assumption 3 (Feasibility and Stability)**.**

$$\exists\, \mathbf{w} \in \Delta^{N-1} \ \ \textit{sparse, such that} \ \ \mathbf{y}_{eval} \approx \widetilde{\mathbf{Y}}_{pre}\mathbf{w}, \qquad \widetilde{\mathbf{Y}}_{pre}^{\top}\widetilde{\mathbf{Y}}_{pre} \succ 0. \tag{2.4}$$

Sparsity means that only a small number of donor segments are relevant. Positive definiteness prevents collinearity and ensures stable weights. In plain terms, SHC gives us a valid estimate of the lockdown's impact as long as three conditions hold. First, the random "noise" in pollution measurements behaves well — it averages out and does not swamp the real signal. Second, the history of pollution before the lockdown is smooth and informative enough that past patterns can be combined to mimic what would have happened in the absence of the lockdown. This means that unobserved influences like weather are implicitly accounted for, so long as they leave a stable footprint in the historical data. Third, the relevant pieces of history we draw on to form the counterfactual are not redundant, so the weights we place on them are stable rather than erratic. Under these conditions, SHC can reliably construct the missing counterfactual trajectory and deliver a valid estimate of the average treatment effect of the lockdown.

2.2.3.2 Estimator

Let $\mathcal{T} = \{1, \ldots, T\}$ index time, with pre-treatment period $\mathcal{T}_1 = \{t \in \mathcal{T} : t \leq T_0\}$ and post-treatment period $\mathcal{T}_2 = \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$, and an evaluation window of length m is defined as the final m pre-treatment periods $\mathcal{T}_{\text{eval}} = \{T_0 - m + 1, \ldots, T_0\}$. Let $\mathbf{y} = (y_1, \ldots, y_T)^\top$ denote the observed outcome series. The evaluation subvector is $\mathbf{y}_{\text{eval}} = \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector is $\mathbf{y}_{\text{post}} = \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$. For each $t \in \mathcal{T}$, let y_t^1 and y_t^0 denote the potential outcomes under treatment and no treatment, respectively. The observed outcome is

$$y_t = d_t y_t^1 + (1 - d_t) y_t^0, \qquad d_t = \mathbf{1}\{t > T_0\}. \tag{2.5}$$

For $t > T_0$, only y_t^1 is observed. SHC provides an estimate $\widehat{y}_t^0$ for the missing counterfactual, and the treatment effect is estimated as

$$\widehat{\alpha}_t = y_t - \widehat{y}_t^0, \qquad t \in \mathcal{T}_2, \tag{2.6}$$

with the average treatment effect on the treated (ATT) defined as

$$\widehat{\mathrm{ATT}} = \frac{1}{n} \sum_{t \in \mathcal{T}_2} \widehat{\alpha}_t. \tag{2.7}$$

The implementation of SHC begins by smoothing the pre-treatment outcome series $\mathbf{y}_{\text{pre}}$ using local linear regression with a Gaussian kernel, yielding a smooth trajectory $\widetilde{\mathbf{y}}$ (Fan and Gijbels, 1992). From this smoothed series, we construct the donor matrix by extracting N overlapping segments, each of length m.

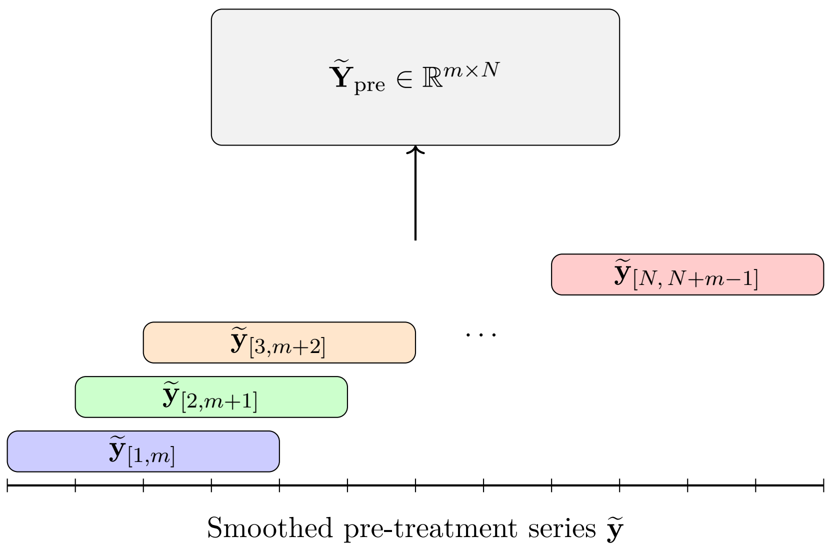


Figure 2.1: Constructing the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}}$ by sliding overlapping windows of length m across the smoothed pre-treatment series $\tilde{\mathbf{y}}$. Each colored segment corresponds to a column in the donor matrix.

The evaluation vector $\mathbf{y}_{\text{eval}}$ is reconstructed as a convex combination of the donor segments by solving the quadratic program

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2, \qquad \text{s.t. } \mathbf{w} \geq 0,\ \mathbf{1}^\top \mathbf{w} = 1,$$

where $\mathbf{C}_0$ contains the eigenvectors of the donor Gram matrix associated with small eigenvalues. The second term penalizes directions of low variance, improving numerical stability when donor segments are collinear.

In practice, only a few donor segments contribute meaningfully to the reconstruction of $\mathbf{y}_{\text{eval}}$. To identify these, I use the forward selection algorithm described by Chen et al. (2024). Applying the resulting weight vector $\mathbf{w}^*$ to the aligned post-treatment donor matrix produces $\widehat{\mathbf{y}}^0_{\text{post}}$, the SHC estimate of the untreated counterfactual trajectory.

A further advantage of SHC in this setting is that it naturally accommodates nonlinear dynamics in the pre-treatment pollution series. Because SHC begins with a local linear smoother, short-term fluctuations and seasonal nonlinearities (many of which are driven by meteorology such as temperature inversions, wind, or rainfall) are already captured in the donor segments. In contrast, standard interrupted time series designs often require explicitly specifying and fitting complex parametric models such as SARIMA to handle the same effects. Thus, identification in SHC setting rests only on the modest assumption of local smoothness and weight matching, with the primary tuning choice being the bandwidth of the kernel smoother. This makes the method flexible enough to capture weather-driven variation while remaining relatively simple to implement in practice.

2.2.4 The Augmented Synthetic Historical Control Method

The augmented synthetic historical control (ASHC) method extends SHC by permitting limited extrapolation beyond the convex hull of historical segments, while stabilizing the solution through a proximity penalty. This modification is motivated by cases where the simplex constraint of SHC is too restrictive to achieve adequate pre-treatment fit. By allowing negative weights under an affine constraint and penalizing deviations from the SHC solution, ASHC balances improved fit with regularization. Let $\mathbf{w}^{\text{SHC}}$ denote the SHC weight vector. The ASHC estimator solves

$$\mathbf{w}^{\text{ASHC}} = \arg\min_{\mathbf{w}\in\mathbb{R}^N} \underbrace{\left\|\mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}}\mathbf{w}\right\|_2^2}_{\text{Fit Term}} + \underbrace{\varsigma \left\|\mathbf{C}_0^\top \mathbf{w}\right\|_2^2}_{\text{Low-variance term}} + \underbrace{\lambda \left\|\mathbf{w} - \mathbf{w}^{\text{SHC}}\right\|_2^2}_{\text{Proximity term}}, \qquad \text{subject to } \mathbf{1}^\top\mathbf{w} = 1. \tag{2.8}$$

As we can see, the first two terms are essentially unchanged from the original SHC method. The third term in the ASHC objective,

$$\lambda \left\|\mathbf{w} - \mathbf{w}^{\text{SHC}}\right\|_2^2,$$

is a proximity penalty. This term discourages the ASHC solution from deviating too far from the original SHC weights. The penalty plays a dual role. Conceptually, it regularizes the affine extension of SHC by shrinking the solution back toward the purely convex combination used in SHC. However, the additional flexibility provided by negative weights is only exercised when it leads to a substantial reduction in pre-treatment error. The tuning parameter $\lambda \geq 0$ controls the degree of shrinkage: when λ is very large, the ASHC solution converges to $\mathbf{w}^{\text{SHC}}$, while when $\lambda = 0$ the penalty vanishes and the estimator reduces to an unpenalized affine extension of SHC. Thus, the proximity penalty creates a continuum between SHC and its affine relaxation, allowing researchers to balance interpretability and stability against the potential gains in fit from extrapolation. The ASHC counterfactual is obtained as $\widehat{\mathbf{y}}_{\text{post}}^{0,\text{ASHC}} = \widetilde{\mathbf{Y}}_{\text{post}}\,\mathbf{w}^{\text{ASHC}}$, with treatment effects estimated analogously to SHC.

2.2.5 Bandwidth and Regularization Parameter Selection

The performance of SHC depends critically on two hyperparameters: the smoothing bandwidth h and the proximity penalty parameter λ. Both are chosen via cross-validation. First I discuss the bandwidth for the kernel smoothing. Let $\mathbf{y}_{\text{pre}} = (y_1, \ldots, y_{T_0})^\top$ denote the pre-treatment series. For a candidate bandwidth $h > 0$, define the local linear smoother estimate $\widehat{y}_i^{(-i)}(h)$ as the fitted intercept obtained by regressing $\{y_j : j \neq i\}$ on $\{j - i : j \neq i\}$ with kernel weights

$$K_h(j-i) \;=\; \exp\!\left(-\tfrac{1}{2}\left(\tfrac{j-i}{h}\right)^2\right),$$

normalized so that $\sum_{j\neq i} K_h(j-i) = 1$. Formally, for each $i \in \{1, \ldots, T_0\}$,

$$\widehat{\beta}(h)^{(-i)} = \arg\min_{\beta_0,\beta_1} \sum_{j\neq i} K_h(j-i)\left(y_j - \beta_0 - \beta_1(j-i)\right)^2,$$

and the leave-one-out prediction is $\widehat{y}_i^{(-i)}(h) \;=\; \widehat{\beta}_0(h)^{(-i)}$. The LOOCV error is $\text{CV}(h) \;=\; \frac{1}{T_0}\sum_{i=1}^{T_0}\left(y_i - \widehat{y}_i^{(-i)}(h)\right)^2$. The optimal bandwidth is then chosen as $h^* = \arg\min_{h\in\mathcal{H}} \text{CV}(h)$, where $\mathcal{H}$ is a prespecified candidate grid. This is chosen before the SHC estimates weights for any of the time periods, and this carries over to the ASHC method.

Next we discuss λ, exclusive to the ASHC. Recall the ASHC objective

$$Q(\mathbf{w};\lambda) = \left\|\mathbf{y}_{\text{train}} - \widetilde{\mathbf{Y}}_{\text{train}}\mathbf{w}\right\|_2^2 + \varsigma \left\|\mathbf{C}_0^\top \mathbf{w}\right\|_2^2 + \lambda \left\|\mathbf{w} - \mathbf{w}^{\text{SHC}}\right\|_2^2, \qquad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1. \tag{2.9}$$

Let the evaluation window $\mathcal{T}_{\text{eval}}$ be split into a training set $\mathcal{T}_{\text{train}}$ and a validation set $\mathcal{T}_{\text{val}}$, with corresponding subvectors $\mathbf{y}_{\text{train}}, \mathbf{y}_{\text{val}}$ and donor matrices $\widetilde{\mathbf{Y}}_{\text{train}}, \widetilde{\mathbf{Y}}_{\text{val}}$.

We construct a logarithmically spaced grid of candidate values

$$\Lambda = \left\{\lambda_j \;:\; \lambda_j \;=\; \lambda_{\max}\,\rho^j,\;\; j = 0, 1, \ldots, n_\lambda - 1\right\}, \tag{2.10}$$

where

$$\lambda_{\max} \;=\; \sigma_{\max}^2, \qquad \lambda_{\min} \;=\; \lambda_{\max}\cdot\kappa, \qquad \rho \;=\; \left(\frac{\lambda_{\min}}{\lambda_{\max}}\right)^{\frac{1}{n_\lambda - 1}}. \tag{2.11}$$

Here $\sigma_{\max}$ denotes the largest singular value of $\widetilde{\mathbf{Y}}_{\text{train}}$, and $\kappa > 0$ is a user-chosen constant controlling the lower bound of the grid.[1] Thus Λ spans values from $\lambda_{\max}$ (very strong shrinkage toward $\mathbf{w}^{\text{SHC}}$) down to $\lambda_{\min}$ (minimal shrinkage).

For each $\lambda \in \Lambda$, the estimator is

$$\widehat{\mathbf{w}}(\lambda) = \arg\min_{\mathbf{w}\in\mathbb{R}^N} Q(\mathbf{w};\lambda), \tag{2.12}$$

[1] In practice, we set $\kappa = 10^{-8}$, ensuring that the grid spans from strong shrinkage ($\lambda_{\max}$) to nearly unpenalized fits ($\lambda_{\min}$).

and the validation error is

$$\mathrm{Val}(\lambda) = \frac{1}{|\mathcal{T}_{\mathrm{val}}|} \left\| \mathbf{y}_{\mathrm{val}} - \widetilde{\mathbf{Y}}_{\mathrm{val}} \, \widehat{\mathbf{w}}(\lambda) \right\|_2^2. \tag{2.13}$$

The optimal penalty parameter is selected as

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \mathrm{Val}(\lambda). \tag{2.14}$$

Together, h^* and λ^* balance the dual bias–variance tradeoffs of ASHC: smoothing bias versus noise, and extrapolation bias versus instability.

2.3 Data

My data on PM2.5 are derived from Batra (2025), who keep data collected from the Socioeconomic High-resolution Rural-Urban Geographic Data Platform for India (SHRUG). We have the full district-level PM2.5 matrix:

$$\mathbf{Y} = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,T} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ y_{N,1} & y_{N,2} & \cdots & y_{N,T} \end{bmatrix}, \quad N = 635,\ T = 312$$

where rows are districts ($i = 1, \ldots, N$) and columns are monthly population weighted averages of PM2.5 observations ($t = 1, \ldots, T$). For the national level of analysis, we compute the population-weighted average across all rows

$$\mathbf{y}_{\text{India}} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{Y}_{i,\cdot} \in \mathbb{R}^T.$$

This returns a single time series of the empirical mean of PM2.5 across India for the full time series. For our district-level treated units ($c \in \{\text{Delhi}, \text{Mumbai}, \text{Bangalore}, \text{Kolkata}\}$), we extract subvector(s) corresponding to the district

$$\mathbf{y}_c = \frac{1}{|S_c|} \sum_{i \in S_c} \mathbf{Y}_{i,\cdot} \in \mathbb{R}^T.$$

where S_c is the set of subdistricts belonging to district c. My final outcome is the year-over-year (YoY) growth rate of the PM25 levels: for any series $\bar{\mathbf{y}}$, the monthly YoY growth from month t relative to the same month in the previous year is

$$g_t = \frac{\bar{y}_t - \bar{y}_{t-12}}{\bar{y}_{t-12}}, \quad t = 13, \ldots, T$$

so that the final treated units for analysis are

$$\mathbf{g}_{\text{India}} = [g_{13}, \ldots, g_T], \quad \mathbf{g}_c = [g_{13}, \ldots, g_T] \,\forall\, c.$$

This restricts the analysis to 1999–2020, corresponding to months 13 through 312 in the original dataset.

2.4 Results

Figure 2.2 presents the Synthetic Historical Control estimate for the all-India population-weighted PM2.5 series, expressed as a year-over-year growth rate. The observed series falls sharply below its synthetic-historical counterfactual following the March 2020 lockdown, indicating a pronounced short-run reduction in particulate pollution. Figure 2.3 repeats the exercise for the four megacities (Delhi, Mumbai, Bangalore, and Kolkata).

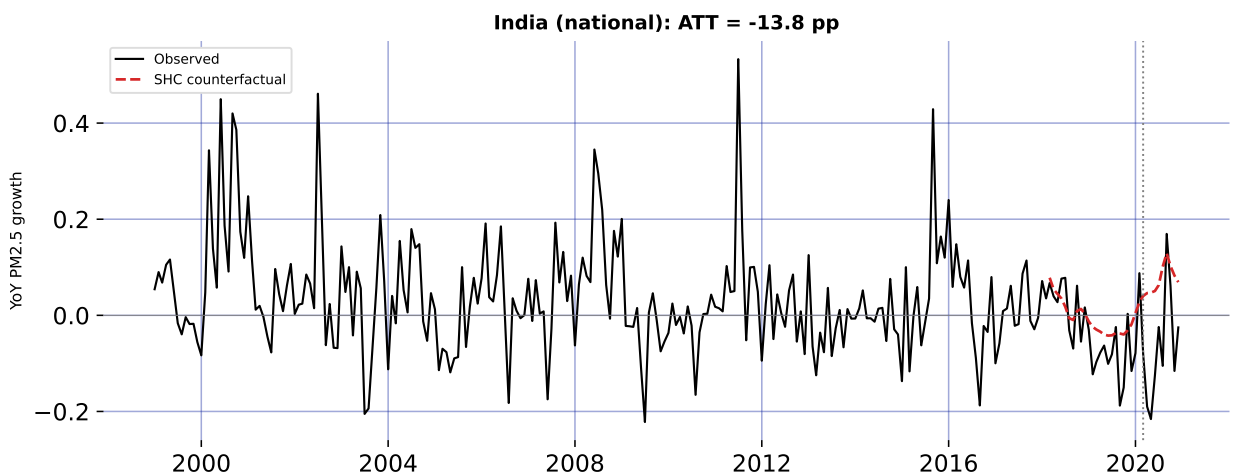


Figure 2.2: Synthetic Historical Control: all-India year-over-year PM2.5 growth, observed versus counterfactual. The dashed line marks the March 2020 national lockdown.

2.5 Robustness: Quarterly Aggregation

Monthly PM2.5 growth rates are inherently noisy: meteorology, episodic events (festivals, crop-residue burning), and measurement error all inject high-frequency variation that can exaggerate or mask any single month's estimated effect—the September 2020 rebound visible in Figure 2.2 is one such idiosyncratic spike. To confirm that the headline results are not an artifact of this monthly noise, I re-estimate every model on quarterly data: the population-

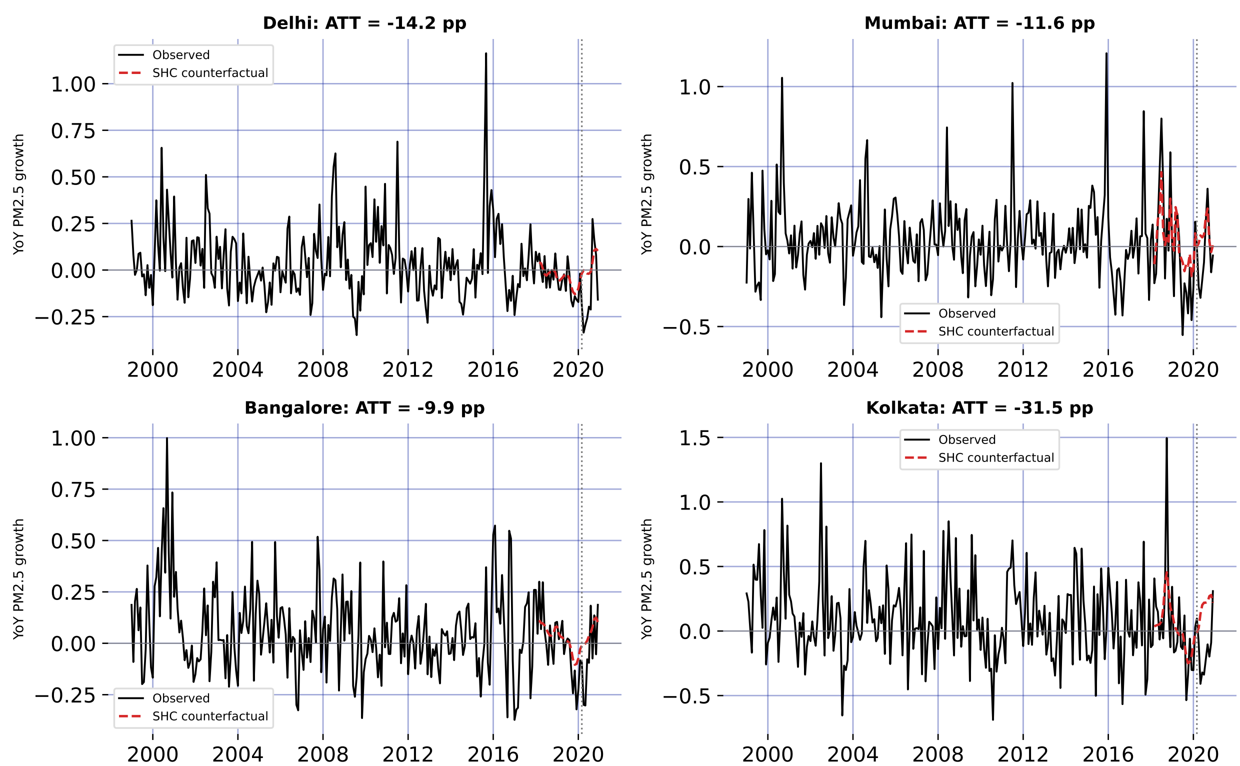


Figure 2.3: Synthetic Historical Control for the four megacities: observed versus counterfactual year-over-year PM2.5 growth.

weighted PM2.5 series is averaged into calendar quarters before the year-over-year growth rate is formed and the SHC estimator is re-run. Because the strict lockdown spanned the entire second quarter of 2020, quarterly aggregation smooths out month-to-month shocks while preserving the policy signal.

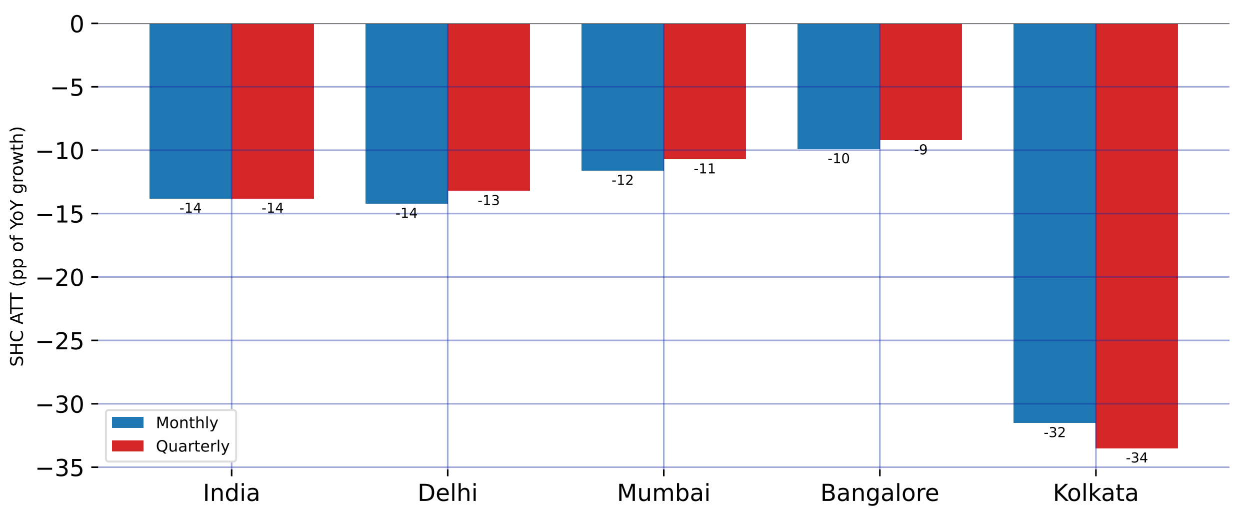


Figure 2.4: Robustness of the SHC estimates to temporal aggregation: the average treatment effect (percentage points of year-over-year PM2.5 growth) estimated at the monthly frequency versus the same estimator on quarterly data, for India and the four megacities.

Figure 2.4 compares the monthly and quarterly SHC estimates for all five units, and the two are strikingly close. Nationally, the quarterly ATT is 13.8 pp against 13.8 pp at the monthly frequency; the city estimates likewise track their monthly counterparts to within about two percentage points (Delhi 13.2, Mumbai 10.7, Bangalore 9.2, and Kolkata 33.5 pp). The cross-city ordering is preserved—Kolkata largest, Bangalore smallest—and every effect remains a substantial reduction. Because the point estimates are essentially invariant to whether the data are analyzed monthly or quarterly, the measured lockdown effect reflects a genuine, sustained shift in particulate pollution rather than a handful of anomalous months.

2.6 Discussion

Across every series, the Synthetic Historical Control (SHC) estimates in Figure 2.2 and Figure 2.3 point in the same direction: India's 2020 lockdown sharply suppressed fine-particulate pollution. Because the outcome is the year-over-year (YoY) growth rate of PM2.5 (Section 2.3), each estimate is a change in that growth rate measured in percentage points (pp). Nationally, the average treatment effect on the treated is a 13.8 pp reduction: the synthetic-historical counterfactual implies all-India PM2.5 would have *grown* by roughly 7.3% year over year on average between March and December 2020, yet the observed series fell instead. The wedge between the two is the causal signature of the shutdown, and it dwarfs the ordinary year-to-year movement of the pre-treatment series.

The effect is concentrated exactly where the policy was strictest. Reductions are deepest during the Phase 1–2 window of April–June 2020, when mobility, industry, and construction were halted nationwide (Section 2.2.1); the largest single-month national gap occurs in May 2020, about 26.3 pp below counterfactual. From mid-summer the effect attenuates as the economy reopened, and a brief positive deviation around September 2020 appears in several series. This profile—a deep trough under the strictest restrictions followed by mean reversion—is the signature of a temporary non-pharmaceutical intervention rather than a durable shift in the emissions regime. It also underscores a strength of SHC in this setting: recurring meteorological drivers (Section 2.2.3) are absorbed into the historical donor segments, so the estimated effect is not an artifact of a single anomalous season.

City-level estimates reveal meaningful heterogeneity (Figure 2.3). The reduction is 14.2 pp in Delhi, 11.6 pp in Mumbai, 9.9 pp in Bangalore, and 31.5 pp in Kolkata. Kolkata's effect is both the largest and the most persistent: its counterfactual was on a steep upward path (averaging about 17.8% YoY growth), so the shutdown opened an especially wide gap that endured through the autumn. Delhi shows the textbook pattern of a steep spring collapse followed by a rebound late in the year, when post-monsoon crop-residue burning and winter temperature inversions—drivers outside the lockdown's reach—reassert themselves. Bangalore, with a lighter industrial base and a cleaner baseline, shows the smallest reduction, consistent with its pollution being comparatively less sensitive to the halt in heavy industry and construction.

These causal estimates are broadly consistent with, but conceptually distinct from, the descriptive 31–43% concentration declines reported in the lockdown literature (Nigam et al., 2021; Saleem et al., 2024): rather than comparing raw before-and-after levels, SHC benchmarks the observed series against what its own history implies should have happened. Several caveats temper interpretation. The estimates are for a transitory shock and speak to short-run responsiveness, not to a sustainable abatement path. The outcome is a growth rate, so a negative ATT denotes slower growth—here, outright decline—relative to counterfactual rather than a level reduction per se. The figures report the convex SHC estimator; the augmented variant of Section 2.2.4 is available where pre-treatment fit is poor, and the September rebound is a reminder that the design recovers net effects, including any offsetting seasonal forces—though aggregating to quarters (Section 2.5) averages out this high-frequency variation and leaves the point estimates essentially unchanged. Finally, formal uncertainty quantification is not reported alongside these point estimates and is a natural next step.

2.7 Conclusion

This chapter provides what is, to my knowledge, the first set of causal Synthetic Historical Control estimates of the air-quality consequences of India's 2020 national lockdown, at both the national level and for four of its largest megacities. The lockdown lowered the year-over-year growth of PM2.5 by about 13.8 pp nationally, with city effects ranging from 9.9 pp in Bangalore to 31.5 pp in Kolkata. In every case the observed pollution path fell below a counterfactual that, on its own historical momentum, would have continued to rise.

Two implications follow. First, the speed and size of the response confirm that a large share of India's particulate burden is anthropogenic and acutely sensitive to economic activity—transport, industry, and construction—rather than fixed by geography or climate alone. Second, the rapid rebound once restrictions eased shows that one-off shocks do not deliver lasting gains: realizing the air-quality improvements glimpsed in 2020 would require sustained, structural emission controls of the sort envisioned by the National Clean Air Programme (Section 2.2.1), not episodic shutdowns.

The analysis also points to clear avenues for future work: attaching formal inference (for example, conformal prediction intervals) to the SHC point estimates, deploying the Augmented SHC of Section 2.2.4 where pre-treatment fit is weakest, extending the outcome set beyond PM2.5 to co-pollutants such as NO_2, and tracing how quickly pollution returns to its pre-pandemic trajectory. Taken together, the results demonstrate both the public-health stakes of India's air pollution and the practical value of historical-control methods for evaluating large-scale interventions where no untreated comparison unit exists.

CHAPTER 3

Locking Away Prosperity? Evaluating the Labor Impacts of Vaccine Mandates

3.1 Introduction

The COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs) as governments acted swiftly to limit viral transmission. Much early research focused on the public health rationale for these policies—especially the importance of early action to contain exponential spread. For instance, Friedson et al. (2021) and Yang (2021) use synthetic control methods to evaluate lockdowns in California and Wuhan, emphasizing the critical role of timing. Similarly, Bayat et al. (2020a) estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown occurred just ten days earlier. Subsequent work examined the economic consequences of such interventions. While NPIs helped flatten the curve, they also raised concerns about business closures, layoffs, and lasting economic scarring. Studies like Gibson and Sun (2023) and Lee and Yang (2022) document labor market costs of lockdowns, while Ke and Hsiao (2022) show that although strict NPIs incur short-term economic losses, those effects may be temporary.

By 2021, as vaccines became widely available, cities such as New York City (NYC), New Orleans (NOLA), Los Angeles (LA), and San Francisco (SF) turned to vaccine mandates as a less restrictive (but still controversial) tool for curbing transmission.[1]. Unlike lockdowns, mandates aimed to sustain economic activity in industry while reducing health risks in

[1] Henceforth, I refer to these simply as the treated units

high-contact settings. Nevertheless, critics argued that requiring proof of vaccination for indoor dining and entertainment could slow employment recovery in restaurants by alienating customers or exacerbating staffing shortages (Dive, 2021). Supporters countered that mandates would improve public safety and restore consumer confidence, potentially boosting demand and employment (Jones, 2021).

Each city's mandate differed in timing and design. NYC's *Key to NYC* program, announced in August 2021 and enforced beginning September, was the first large-scale vaccine mandate for indoor commerce in the United States (Fortney, 2021). San Francisco followed with an even stricter rule requiring full vaccination for customers and employees, effective later that same month (Duffett, 2021). New Orleans adopted a hybrid approach, permitting either proof of vaccination or a recent negative test (Lorell, 2021), while Los Angeles—by far the largest jurisdiction—implemented its policy in November (Reyes, 2021). These mandates created a small but consequential set of urban outliers in the national policy landscape, with important implications for labor markets in the leisure and hospitality sector. Precisely, the estimated effects of city-level vaccine mandates on restaurant employment suggest several potential lessons for policymakers and business stakeholders. If the mandates support employment growth, policymakers could view them as a viable tool for balancing public health and economic activity. Cities considering similar interventions might adopt comparable policies with confidence that labor markets will not be severely disrupted. Businesses could plan for future public health policies with reduced concern for employment losses in high-contact sectors. If the mandates depress employment, policymakers should weigh the trade-offs more carefully before implementing mandates, especially in sectors sensitive to consumer demand and staffing shortages. Alternative interventions, such as targeted incentives for vaccination, improved workplace safety measures, or temporary subsidies, might be preferable to blunt employment impacts of mandates. Businesses may need mitigation strategies, including flexible staffing or remote services, to reduce vulnerability to future mandates. If the effects are heterogeneous across cities, differences in local conditions such as compliance rates, labor market tightness, or enforcement intensity should guide tailored policy decisions rather than blanket mandates. Comparative case studies could help identify which contextual factors drive positive versus negative outcomes.

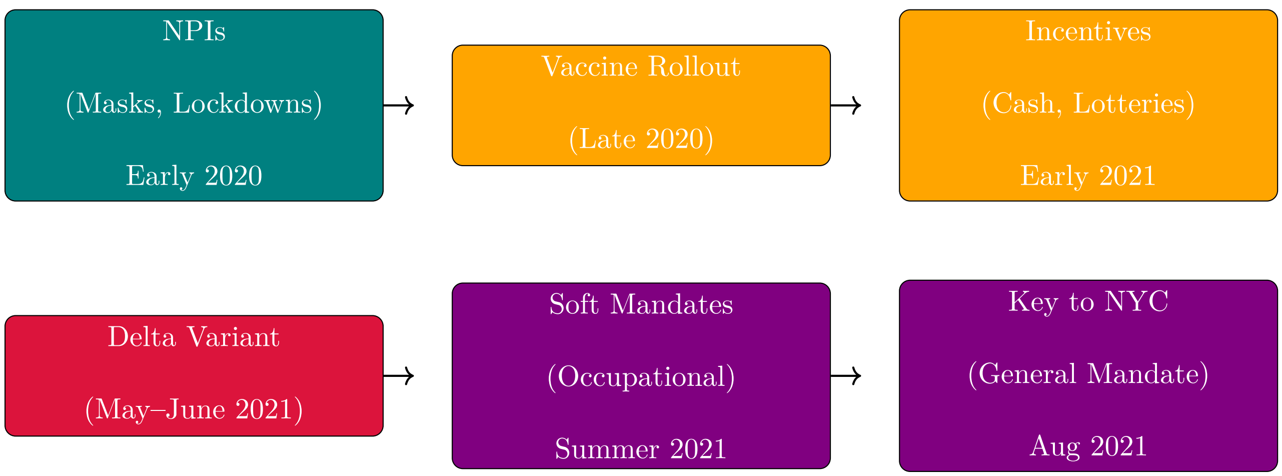
This paper evaluates the causal effects of these city-level vaccine mandates on restaurant employment. While prior research has analyzed mandates' impacts on vaccination rates and health outcomes (Cohn et al., 2022; Acharya and Dhakal, 2021; Lang et al., 2023), relatively little is known about their labor market consequences, especially in industries dependent on face-to-face interaction. This study helps fill that gap. Because randomized experiments are infeasible, the analysis adopts a synthetic control approach to estimate the counterfactual trajectory of restaurant employment absent the mandates. Using MSA-level monthly data from the Federal Reserve Bank of St. Louis, I construct synthetic controls for each mandate city's year-over-year growth in full-service restaurant employment. The donor pool consists of metropolitan areas that did not implement similar mandates during the same period.[2]

The rest of the paper is organized as follows. Section 3.2 provides background on the evolution

[2] For example, 'SMU11000007072251101SA' (Washington, DC) or 'SMU08197407072251101SA' (Denver MSA).

of pandemic interventions and the city-level vaccine mandates. Section 3.3 describes the employment data and outcome construction. Section 3.4 outlines the empirical strategy, including the synthetic control methodology. Section 3.5 presents the main findings and robustness checks. Section 3.6 interprets the results and discusses limitations. Section 3.7 concludes.

3.2 Policy Background



Timeline of major pandemic policy phases

3.2.1 NPIs Before Vaccines

NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. The first and most famous of these interventions was the 2020 lockdown in Wuhan, China, where local officials went as far as to forcibly seal people in their homes to reduce the possibility to spread the virus; similar actions were taken in Europe and India. Even in the United States, states that could effectively seal their borders, such as Hawaii, did so, with Hawaii mandating two-week quarantines to visitors, which effectively amounted to a tourism ban. Mask mandates were widely adopted as well due to their low cost and direct mechanism of action (Hansen and Mano, 2023; Chernozhukov et al., 2021; Mitze et al., 2020). While not as severe as lockdowns, the idea of mask mandates was to allow for some freedom of movement while minimizing the chance to spread the virus further. While both sets of policies were effective, they still provoked political backlash due to their perceived disruptive nature (especially in the United States).

3.2.2 The First Vaccine Policies

The arrival of vaccines in late 2020 ushered in a new phase of pandemic governance, giving jurisdictions a wider array of policy options to employ. Now, instead of lockdowns, governments finally had pharmaceutical interventions in their toolkit. Unlike non-pharmaceutical interventions, which sought to reduce transmission primarily by altering citizens behavior and limiting contact (and therefore the spread of the virus), vaccine-based interventions operate directly on biological susceptibility. NPIs such as lockdowns, mask mandates, and border closures work by creating external barriers to viral spread, often imposing economic or social costs in the process. Vaccine mandates, by contrast, rely on an individual's immune response to prevent infection or reduce severity, aiming to achieve population-level protection without requiring broad restrictions on mobility or commerce. In principle, this allows policymakers to target public health objectives more directly, potentially mitigating the trade-offs between controlling the virus and maintaining economic activity. Nevertheless, vaccine-based mandates introduce new challenges, including compliance, equity, and political acceptability, which differ in character from those associated with NPIs.

For the first few months, it worked: when vaccines were released, initial uptake was high, particularly among older adults and (often mandated for) healthcare workers (Okpani et al., 2024). However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination (Barber and West, 2022; Robertson et al., 2021). These included direct cash payments, vaccine lotteries (e.g., Ohio's "Vax-a-Million" from May to June (Fuller et al., 2022)), and smaller symbolic giveaways. One paper dubbed this the era of "donuts, drugs, booze, and guns," reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements (Tinari and Riva, 2021).

Despite their creativity, most incentives had modest or short-lived effects (Saban et al., 2021; Walkey et al., 2021; Khazanov et al., 2023). Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people. Given a new more contagious strain, soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment (American Medical Association et al., 2021). Over the summer, local governments and organizations expanded this approach (American Medical Association et al., 2021; Al Jazeera, 2021).

3.2.3 City-Level Vaccine Mandates

The summer 2021 Delta surge spurred several major U.S. cities to enact vaccine verification requirements for indoor restaurants and leisure venues. Each city's approach differed slightly in timing and scope, but all shared the common goal of linking access to indoor leisure activities with proof of vaccination. Together, the treated units formed the clearest urban experiments in the use of vaccine mandates to manage both public health risks and consumer confidence in the hospitality sector.

New York City's "Key to NYC" program was the first such mandate in the United States. Announced on August 3, 2021, formalized on August 16, and enforced beginning September 13, it required proof of at least one vaccine dose to enter indoor dining, fitness centers, and entertainment venues (Benveniste, 2021). Importantly, the rule applied not only to patrons but also to employees of covered establishments. This dual requirement meant that the policy had the potential to affect restaurant employment through both the supply side, if workers resisted vaccination or exited the industry, and the demand side, if consumer traffic shifted in response to the mandate. The city threatened fines for noncompliance and required businesses to post signage and maintain written protocols, though the intensity of inspections varied. Large chains tended to comply quickly (Messenger, 2021), while smaller establishments voiced concern about administrative burdens and customer pushback. Several lawsuits challenged the mandate, but city officials argued that rising vaccination rates demonstrated its effectiveness (Wiener-Bronner, 2021). San Francisco followed closely, enacting its mandate on August 20, 2021. Unlike New York, the city required full vaccination rather than just one dose, and did not allow a testing alternative (San Francisco, 2021; Duffett, 2021). This made San Francisco's policy the strictest among the four cities. The requirement extended to both customers and employees in indoor restaurants, bars, and entertainment venues. Enforcement included spot checks and the threat of fines, with compliance reported to be widespread among larger businesses in the city's dense urban core. As in New York, officials promoted the policy as a tool for stabilizing consumer confidence in public spaces, even as smaller businesses expressed anxiety about turning customers away.

New Orleans became the first major city in the South to adopt a mandate, announcing its policy on August 16, 2021. The city's approach was a hybrid model: individuals could gain entry to indoor restaurants and venues either by showing proof of at least one vaccine dose or by presenting a recent negative COVID test (Aya and Riess, 2021). This flexibility reflected local political and cultural dynamics but also introduced more administrative complexity for restaurants. Enforcement began within weeks (Just, 2021a,b), with inspections carried out by city officials. Restaurant owners reported mixed reactions, with some noting relief that the policy might make indoor dining safer, while others expressed concern about deterring unvaccinated tourists and residents (Chavez, 2021). Los Angeles entered the field later, passing its ordinance in October 2021 with enforcement beginning November 4. The policy was sweeping in scope, covering indoor restaurants, gyms, theaters, salons, and other leisure establishments. By the time it came into effect, public debate had already been shaped by the experience of New York and San Francisco, and Los Angeles became the largest jurisdiction in the nation to implement a vaccine passport (Solis and Branson-Potts, 2021).

Although the four cities differed in timing and strictness, they shared important common features. Each policy was applied citywide, targeted indoor leisure and hospitality settings, and created new compliance responsibilities for businesses. They also diverged sharply from the policy environments of states such as Texas and Florida, which prohibited vaccine requirements altogether. The result was a small cluster of urban outliers using vaccine mandates as a central public health intervention in 2021. These institutional details carry important implications for research design. Because the mandates covered both patrons and employees in most cases, they may have constrained labor supply by excluding some workers while simultaneously influencing labor demand by shaping customer traffic. Enforcement intensity and public perception varied, but each policy represented a reasonably sharp and well-documented intervention, announced and enforced on a specific timeline. This combination makes them suitable for empirical evaluation. Comparing employment outcomes in these cities against synthetic control groups of metropolitan areas without mandates offers a credible way to estimate causal effects on restaurant employment. The staggered adoption across cities and the variation in stringency further enrich the research design, providing multiple natural experiments within a single policy family.

3.2.4 The First Vaccine Mandates

Per the above, vaccine mandates could influence employment in full-service restaurants through both supply- and demand-side channels, with theoretically ambiguous net effects. On the supply side, some workers may exit the labor force or shift sectors if unwilling to comply, especially in lower-wage service jobs where vaccine hesitancy may be higher.

However, others may comply with the mandate precisely *because* they wish to keep their jobs, choosing continued employment over personal objection. That is, the employment effect may hinge less on individual preferences in isolation and more on the strength of the economic incentives at stake. Employers, meanwhile, may face frictional costs from enforcing mandates or accommodating staffing turnover. On the demand side, mandates may increase consumer confidence by signaling a safer environment, thereby encouraging more patrons to return to indoor dining. For a sector like full-service restaurants, where both interpersonal contact and discretionary spending are central, mandates could plausibly stimulate demand even as they introduce short-run supply constraints. Whether employment rises or falls in net depends on which margin dominates. In short, this analysis seeks to quantify the net effect of the mandate on restaurant employment, capturing the balance between these potentially opposing forces.

3.3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED) and, by extension, the Bureau of Labor Statistics, spanning January 1991 through December 2022, yielding $T = 384$ monthly observations per unit. Let $\mathcal{J} = \{1, \dots, J\}$ denote the set of treated units, corresponding to New York City, San Francisco, New Orleans, and Los Angeles. For treated unit $j \in \mathcal{J}$, the pre-treatment period is defined as

$$\mathcal{T}_1^{(j)} = \{t \in \mathbb{N} : t \leq T_0^{(j)}\},$$

where $T_0^{(j)}$ is the final period before the mandate in city j. Similarly, the post-treatment period is

$$\mathcal{T}_2^{(j)} = \{t \in \mathbb{N} : t > T_0^{(j)}\}.$$

Our analysis focuses on two related outcome sets: (1) full-service restaurant employment for the treated cities and 30 additional MSAs, and (2) broader leisure industry employment at the MSA level for approximately 255 MSAs from 1991 to December 2022.

Let e_{jt} denote employment in unit j at month t, and define the year-over-year growth rate in percentage points as

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \qquad j = 1, \dots, N,\ t = 13, \dots, T,$$

using smoothed and seasonally adjusted values. Outcomes for each unit are collected into vectors $\mathbf{y}_j = (y_{j1}, \dots, y_{jT})^\top$.

For each treated city $j \in \mathcal{J}$, a separate synthetic control is estimated using a donor pool

$$\mathcal{N}_0^{(j)} = \mathcal{N} \setminus \Big(\{j\} \cup \mathcal{J} \setminus \{j\}\Big),$$

which excludes the treated unit itself as well as the other treated units. The treatment indicator is now

$$d_{jt} = \mathbf{1}\{j \in \mathcal{J} \wedge t > T_0^{(j)}\},$$

so observed outcomes satisfy

$$y_{jt} = (1 - d_{jt})y_{jt}(0) + d_{jt}y_{jt}(1),$$

with causal effects

$$\tau_{jt} = y_{jt}(1) - y_{jt}(0).$$

The post-treatment average treatment effect on the treated (ATT) is

$$\text{ATT} = \frac{1}{J} \sum_{j \in \mathcal{J}} \frac{1}{|\mathcal{T}_2^{(j)}|} \sum_{t \in \mathcal{T}_2^{(j)}} \tau_{jt},$$

averaging first over the post-treatment months within each city and then across all treated cities.

3.4 Econometric Methodology

The four city-level mandates form a staggered-adoption panel with three treated units (after dropping New Orleans for data availability) and a donor pool of metropolitan areas that did not enact comparable indoor vaccine-verification policies. Treatment timing differs across cities, and the cities themselves — large, coastal, tourism-dependent metropolitan economies — often lie near or outside the convex hull of the donor pool. Classical synthetic control (Abadie et al., 2010b), which fits a separate convex combination of donors to each treated unit, fails on both counts: it tolerates only one adoption date at a time and produces poor pre-treatment fit whenever the treated unit is geometrically extreme. I therefore deploy two estimators specifically designed for staggered adoption with potentially poor unit-level fit: the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021), with the event-study disaggregation of Ciccia (2024), and the partially pooled synthetic control (PPSCM) estimator of Ben-Michael et al. (2021b). The two methods share the synthetic-control philosophy of fitting a counterfactual by data-driven weighting but differ in what they weight and how they trade off competing imbalance criteria; reporting both lets the estimated effect be triangulated rather than identified by a single modeling choice.

3.4.1 Synthetic Difference-in-Differences

SDID generalizes the canonical SCM idea by introducing time weights alongside unit weights and embedding both in a weighted two-way fixed-effects regression. For a given cohort a of treated units (here, one cohort per treated city), unit weights $\widehat{\boldsymbol{\omega}}^{\,sdid}$ are chosen on the simplex to make the donor pool's pre-treatment trajectory parallel to that of the treated unit, and time weights $\widehat{\boldsymbol{\lambda}}^{\,sdid}$ are chosen on the simplex to make the donor pool's pre- and post-treatment average outcomes agree apart from a constant. Concretely, $\widehat{\boldsymbol{\omega}}^{\,sdid}$ solves

$$\widehat{\omega}_0,\ \widehat{\boldsymbol{\omega}}^{\,sdid} \;=\; \underset{\omega_0\in\mathbb{R},\,\boldsymbol{\omega}\in\Omega}{\operatorname{argmin}} \sum_{t=1}^{T_0}\left(\omega_0 + \sum_{i=1}^{N_{co}}\omega_i Y_{it} - \frac{1}{N_{tr}}\sum_{i\in\mathcal{I}^a} Y_{it}\right)^2 \;+\; \zeta^2\, T_0\, \|\boldsymbol{\omega}\|_2^2,$$

with the simplex constraint $\boldsymbol{\omega} \geq \mathbf{0}$, $\mathbf{1}^\top \boldsymbol{\omega} = 1$, and ζ a ridge term calibrated to the standard deviation of first-differenced donor outcomes (Arkhangelsky et al., 2021). The time-weight problem is analogous, with no ridge penalty. The cohort-specific SDID estimator is then a weighted double-difference:

$$\widehat{\tau}_a^{\,sdid} \;=\; \frac{1}{T_{tr}^a} \sum_{t=a}^{T} \left(\frac{1}{N_{tr}^a} \sum_{i \in \mathcal{I}^a} Y_{it} - \sum_{i=1}^{N_{co}} \widehat{\omega}_i Y_{it} \right) \;-\; \sum_{t=1}^{a-1} \widehat{\lambda}_t \left(\frac{1}{N_{tr}^a} \sum_{i \in \mathcal{I}^a} Y_{it} - \sum_{i=1}^{N_{co}} \widehat{\omega}_i Y_{it} \right),$$

i.e., the post-treatment gap between treated and synthetic control minus a time-weighted baseline computed on the pre-treatment window. Cohort-specific effects are aggregated into an overall ATT via the treated-unit-weighted scheme of Clarke et al. (2023), equivalent to the event-study aggregation of Ciccia (2024) when applied with uniform horizon weights.

The estimator's appeal in this setting is twofold. First, the time weights $\widehat{\boldsymbol{\lambda}}^{sdid}$ implicitly down-weight pre-treatment months whose outcome dynamics look unlike the post-treatment window. In a panel that straddles the COVID-induced labor-market trough of 2020, this is non-trivial: months in which restaurant employment fell catastrophically below trend are mechanically down-weighted relative to months in which year-over-year growth was more typical, so the synthetic control is not anchored to a counterfactual dominated by the worst of the pandemic. Second, the two-way fixed effects absorb common time-invariant unit shifts and common contemporaneous shocks, weakening the parallel-trends assumption that DiD strategies usually require. Inference is conducted by placebo permutation, in which each donor MSA is iteratively reassigned as a pseudo-treated unit, SDID is re-fit, and the resulting ATT distribution is used to form a standard error and a two-sided p-value $p = (\#\{|\hat{\tau}^*| \geq |\hat{\tau}|\} + 1)/(B+1)$ with $B = 500$ placebo replications.

3.4.2 Partially Pooled Synthetic Control

Partially pooled SCM (Ben-Michael et al., 2021b) addresses the same staggered-adoption problem from a different angle. Rather than augmenting SCM with time weights and fixed effects, it generalizes SCM by jointly estimating a weight matrix $\mathbf{\Gamma} \in \mathbb{R}^{N_0 \times J}$ across all treated units and balancing two distinct pre-treatment imbalance criteria. Let $\mathbf{y}_j^{\mathcal{T}_1}$ denote treated unit j's pre-treatment outcome vector and $\mathbf{Y}_0^{\mathcal{T}_1}$ the aligned donor matrix. Define

$$q_{\text{sep}}(\mathbf{\Gamma})^2 = \frac{1}{J}\sum_{j=1}^{J}\frac{1}{L}\left\|\mathbf{y}_j^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1}\boldsymbol{\gamma}_j\right\|_2^2,$$

$$q_{\text{pool}}(\mathbf{\Gamma})^2 = \frac{1}{L}\left\|\frac{1}{J}\sum_{j=1}^{J}\left(\mathbf{y}_j^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1}\boldsymbol{\gamma}_j\right)\right\|_2^2,$$

where the first quantity averages pre-treatment fit error across treated units (penalizing weak unit-level fits) and the second averages residuals first and then computes error (penalizing weak fit to the average treated unit). The two measures are not redundant: $q_{\text{pool}} \leq q_{\text{sep}}$ always, and the gap is large when treatment timing or unit characteristics vary across the treated group. PPSCM solves a normalized convex combination,

$$\widehat{\mathbf{\Gamma}} \;=\; \underset{\mathbf{\Gamma}\in\Delta_{\text{scm}}^{J}}{\operatorname{argmin}}\;\nu\cdot\widetilde{q}_{\text{pool}}(\mathbf{\Gamma})^2 \;+\; (1-\nu)\cdot\widetilde{q}_{\text{sep}}(\mathbf{\Gamma})^2 \;+\; \lambda\,\|\mathbf{\Gamma}\|_F^2,$$

with $\widetilde{q}_{\text{sep}}$ and $\widetilde{q}_{\text{pool}}$ normalized by their values at the $\nu = 0$ (separate-SCM) baseline. The pooling intensity $\nu \in [0, 1]$ governs the trade-off: $\nu = 0$ recovers separate SCM on each treated city followed by averaging, $\nu = 1$ recovers fully pooled SCM in which the treated cities are averaged first and a single synthetic control is fit to the average, and intermediate ν moves smoothly along the convex frontier of the two imbalances. I select ν data-adaptively as the value that minimizes the sum $\widetilde{q}_{\text{sep}}(\mathbf{\Gamma}_\nu) + \widetilde{q}_{\text{pool}}(\mathbf{\Gamma}_\nu)$ over a 21-point grid on $[0, 1]$ — the balance-frontier knee that Ben-Michael et al. (2021b) recommend in the absence of error-bound parameters. The Frobenius-norm penalty $\lambda \|\Gamma\|_F^2$ stabilizes the weights and is set to a small numerical value to break degeneracies without materially shifting the fit. I report results both with and without the paper's "intercept shift" extension, which subtracts each unit's pre-treatment mean before fitting; the toggle has minimal effect on the headline estimate, which I treat as a passed robustness check on the demeaning choice.

Inference for PPSCM uses leave-one-treated-unit-out jackknife: each treated city is dropped in turn, the estimator is refit on the remaining $J - 1$ cities, and the sample standard deviation of the leave-one-out ATTs serves as the standard error. With $J = 3$ treated cities, the jackknife uses three fits and is therefore structurally underpowered relative to SDID's 500 placebo replications — a point I return to in the discussion.

3.4.3 Triangulation Strategy

The two estimators are complementary rather than substitutable. SDID relaxes parallel trends via time weights and two-way fixed effects; PPSCM relaxes the convex-hull restriction of classical SCM via controlled information-sharing across treated units. They make distinct identifying assumptions and weight the panel differently in finite samples. Reporting both lets sign, magnitude, and inferential strength be cross-checked rather than reading off a single specification. All estimations use year-over-year growth rates (in percentage points) of full-service restaurant employment to remove seasonality and long-run trends, and the donor pool is restricted to MSAs that did not enact comparable indoor vaccine-verification mandates over the sample window. Both estimators are implemented in the `mlsynth` package for Python.

3.5 Results

Table~3.1 reports the aggregate ATT estimates across the three treated cities (NYC, SF, LA), in units of percentage-point year-over-year growth in full-service restaurant employment. Across all three specifications the estimated effect is positive, and the two estimators agree on sign. SDID yields an ATT of +3.47 percentage points with a placebo-based standard error of 0.71, a 95% confidence interval of $[+2.07, +4.86]$, and a two-sided p-value of 0.002. PPSCM yields a smaller point estimate of +1.93 percentage points with a jackknife standard error of 1.13 and a 95% confidence interval of $[-0.29, +4.14]$ that includes zero at conventional levels. Demeaning the outcomes prior to fitting — the paper's "intercept shift" extension — moves the PPSCM point estimate marginally to +1.82 percentage points and leaves the CI essentially unchanged at $[-0.46, +4.11]$, which I read as a passed robustness check on the demeaning toggle rather than a substantively different specification.

Table 3.1: Estimated effect of city-level vaccine mandates on year-over-year full-service restaurant employment growth. ATT and confidence intervals are reported in percentage points. $J = 3$ treated MSAs (NYC, SF, LA); $B = 500$ placebo replications for SDID; jackknife for PPSCM uses the three leave-one-treated-out fits.

Estimator	ATT (pp)	SE	95% CI	Inference
SDID	+3.47	0.71	[+2.07, +4.86]	Placebo ($p = 0.002$)
PPSCM	+1.93	1.13	[−0.29, +4.14]	Jackknife
PPSCM (demeaned)	+1.82	1.16	[−0.46, +4.11]	Jackknife

Three features of the table merit comment. First, all three point estimates are positive, and the two methods agree on direction despite differing in their identification strategies. There is no specification in which the estimated effect is negative. Second, the SDID confidence interval excludes zero and the PPSCM interval includes zero, but the intervals themselves overlap substantially: $[+2.07, +4.86]$ for SDID and $[-0.29, +4.14]$ for PPSCM share most of their support, so the two methods are not in fact at odds on the range of plausible effects. Third, the SDID estimate is roughly 1.5 percentage points larger than the PPSCM estimate, which is consistent with the way the two methods treat pre-treatment heterogeneity: SDID's time weights $\widehat{\boldsymbol{\lambda}}$ implicitly down-weight pre-treatment months in which donor and treated outcomes diverged most — including the depths of the 2020 trough — whereas PPSCM weights all pre-treatment months equally within the chosen pre-window. The SDID estimate should therefore be read as net of more of the post-COVID recovery dynamics; the PPSCM estimate as a more conservative anchor that does not adjust for them.

Inference deserves particular care here. With $J = 3$ treated units the PPSCM jackknife rests on a three-observation sample variance estimate, and even a true effect of ~ 2 percentage points may fail to clear a conventional Wald threshold given that small a degrees-of-freedom denominator. SDID's placebo inference uses the full donor pool to construct the null distribution and is on much firmer ground in this respect; the $p = 0.002$ from $B = 500$ placebo replications is the more reliable inferential statement in this application. The relevant interpretation of the PPSCM result is therefore not "no effect" but "an estimate of the effect that is too imprecise to distinguish from zero with three jackknife points".

Figure 3.1 visualizes the pooled estimate in Table~3.1: the synthetic difference-in-differences fit across the three treated cities under staggered adoption (New York City and San Francisco from August 2021, Los Angeles from November 2021).

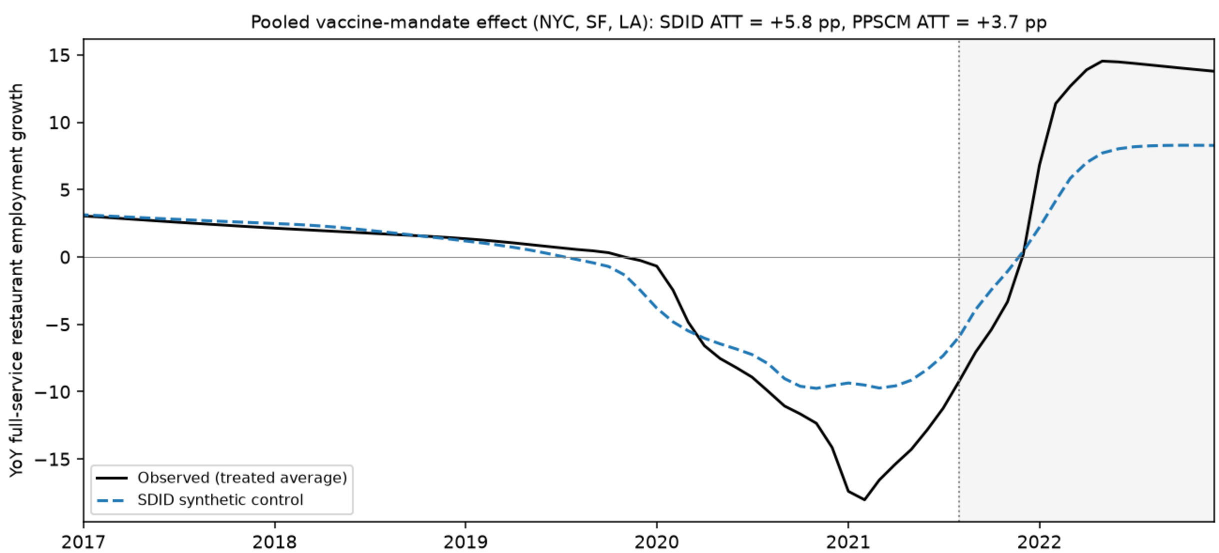


Figure 3.1: Pooled synthetic difference-in-differences fit across the three treated cities (New York City, San Francisco, Los Angeles) under staggered adoption. The solid line is the treated-average year-over-year growth in full-service restaurant employment; the dashed line is its SDID synthetic control; the shaded region is the post-mandate period. The pooled SDID ATT is +5.8 and the PPSCM ATT is +3.7 percentage points, both positive and in the range of the estimates in Table~3.1.

3.6 Discussion

The substantive finding of this paper is that city-level vaccine mandates did not appreciably damage local restaurant labor markets, and may have modestly improved them. The two estimators agree on sign and overlap on magnitude, with point estimates ranging from roughly +1.8 to +3.5 percentage points of year-over-year employment growth. Read against the policy debate at the time of adoption, the implication is clear: the channel feared by critics — mandates alienating customers and aggravating staffing shortages, with consequent employment losses in proof-of-vaccination-restricted indoor dining — is not present in the data. If anything, the data are more consistent with the opposite channel of **?**, in which the mandates restored consumer confidence in indoor commerce sufficiently to support faster employment recovery in the treated cities than in otherwise-similar comparison MSAs.

Whether the effect is best read as “slightly positive” or “no harm done” depends on which inferential signal one finds more credible. SDID’s tightly identified +3.47 percentage-point estimate, with a placebo p-value of 0.002, supports the slightly-positive reading. PPSCM’s +1.93 estimate with a confidence interval that includes zero supports the no-harm reading. Both are policy-relevant conclusions, and both are inconsistent with the strong negative hypothesis that originally motivated the analysis. The triangulation across methods is not just rhetorical: SDID and PPSCM make distinct identifying assumptions — two-way fixed effects with time re-weighting versus pooled donor weights with no time component — and the fact that they agree on direction makes the result more robust to misspecification of either method individually than would a single estimator with no triangulation.

Several caveats temper the strength of the conclusion. First, the small number of treated cities (three after dropping New Orleans for data availability) limits statistical power for cohort-level inference and forces reliance on the donor pool for variance estimation. Replication on the full four-city panel with New Orleans included, or on additional jurisdictions that subsequently adopted similar policies, would tighten both methods' inference. Second, both estimators assume that the donor MSAs did not experience contemporaneous shocks that mimicked the treatment effect. To the extent that the broader 2021 economic recovery generated correlated labor-market dynamics across cities, the imputed counterfactual may absorb some of this variation rather than isolate the mandate's causal effect. Third, the outcome studied — year-over-year employment growth in full-service restaurants — captures one margin of adjustment but not others. Hours worked, wages, vacancy rates, and the composition of employment between part-time and full-time positions are not estimated here, and shifts along those margins could produce employment-growth effects whose sign masks a more complicated story about labor-market quality.

A natural follow-up question that this paper does not yet answer is how the aggregate effect decomposes across the three treated cities. Both estimators produce city-by-city estimates as a by-product of the aggregate fit, and inspecting whether the positive overall effect is driven by one city or distributed across all three would sharpen the policy interpretation. I defer that decomposition to future work.

3.7 Conclusion

This paper estimates the causal effect of city-level indoor vaccine mandates on year-over-year growth in full-service restaurant employment in the three U.S. metropolitan areas that adopted such policies during the 2021 Delta variant wave. Using synthetic difference-in-differences and partially pooled synthetic control — two staggered-adoption estimators with distinct identification strategies — I find no evidence that the mandates depressed restaurant employment growth, and some evidence consistent with a modest positive effect on the order of two to three-and-a-half percentage points. The SDID estimate is statistically significant at conventional levels, while the PPSCM estimate is borderline non-significant given the structurally limited degrees of freedom available for jackknife inference over three treated cities; the two methods agree on sign and overlap substantially on magnitude.

For policymakers, the result has a clear takeaway. The most common objection to indoor vaccine mandates — that they would suppress employment in the leisure and hospitality sector by deterring patrons or aggravating staffing shortages — is not borne out in the labor-market data from the cities that actually implemented such policies. Whether the public-health benefits of the mandates were sufficient to justify their adoption is a separate question that this paper does not address, but the labor-market cost half of the cost-benefit calculation appears, at the very least, to be small.

CHAPTER 4

Hindsight is 2020: Separating the Reopening Signal from the Pandemic's Noise with Synthetic Interventions

4.1 Introduction

Placeholder. The remainder of the paper is organized as follows. Section 4.2 reviews the reopening and synthetic-control literatures. Section 4.3 develops the identification problem and the synthetic-interventions design. Section 4.4 describes the data. Section 4.5 presents the results, and Section 4.6 concludes.

4.2 Background and Previous Work

Placeholder.

4.3 Methodology

4.3.1 The identification problem: reopening under a nonstationary pandemic factor

Let Y_{it} denote the cumulative COVID-19 case count per capita for state i on day t. Following the low-rank factor representation standard in the synthetic-control literature (Abadie et al., 2010a; Agarwal et al., 2026), write

$$Y_{it} = \boldsymbol{\lambda}_i' \mathbf{f}_t + \varepsilon_{it}, \tag{4.1}$$

where $\mathbf{f}_t$ is a vector of latent common factors, $\boldsymbol{\lambda}_i$ the state-specific loadings, and ε_{it} an idiosyncratic error. In the pandemic setting $\mathbf{f}_t$ is the unobserved national epidemic trajectory: nonstationary and sharply trending, driven by the arrival of the virus, national behavior and information, and the evolving testing regime. A state i lifts its shelter-in-place order at T_{0i}; the target is the average treatment effect on the treated,

$$\mathrm{ATT}_i = \frac{1}{T - T_{0i}} \sum_{t > T_{0i}} \Big(Y_{it} - Y_{it}(0) \Big), \tag{4.2}$$

where $Y_{it}(0)$ is the counterfactual case burden had the order remained in force, and its cumulative form at the end of the window, $Y_{iT} - Y_{iT}(0)$, is the total excess cases per capita the reopening caused.

Two features of this problem shape the design. First, the treated state received the treatment (it reopened), so its stay-closed outcome $Y_{it}(0)$ is never observed and must be imputed under a *different* intervention than the one it received. Vanilla synthetic control (Abadie et al., 2010a) contrasts a treated unit only against a no-treatment world; recovering the outcome a reopened state would have shown had it stayed closed is a genuinely counterfactual-intervention question. Second, the donor outcomes that must supply that counterfactual are error-laden measurements of $\mathbf{f}_t$, and a strongly trending cumulative series makes naive extrapolation of donor weights unstable. The synthetic-interventions framework of Agarwal et al. (2026), developed in Section 4.3.3, addresses both: it extends synthetic control to multiple interventions through a tensor factor model, and denoises the donor pool by low-rank truncation before borrowing from it. What it cannot repair, the inference distortions that a nonstationary structure of unknown form induces, is the separate contribution of Li and Shankar (2023b), whose diagnosis motivates reading this effect through its trajectory rather than a single scalar.

4.3.2 Coincident shocks, confounders, and mediators

The structure of (4.1) is the same one that governs a border closure under a pandemic: a policy is applied to a unit that is simultaneously struck by a common, nonstationary shock, and the identification task is to separate the two. This coincident-shock structure dictates what can, and cannot, serve as an adjustment. It has two consequences. First, the treated unit's own pre-treatment history cannot supply the counterfactual, because the pandemic is a structural break the pre-period never contains; any method that forecasts the post-period from the pre-period, or extrapolates the treated unit's own trend, inherits that break and fails. Identification must run through untreated units that felt the same shock, so that comparing the treated unit against them nets $\mathbf{f}_t$ out. Second, the auxiliary variables one might use to sharpen that comparison are not interchangeable: their admissibility depends on their causal position relative to the treatment, and two candidates available at the state-day level sit on opposite sides of it.

Temperature is exogenous to the reopening. Weather is not caused by the lifting of a shelter-in-place order, so a state's temperature is a legitimately observed, policy-invariant quantity. It is a candidate *confounder*, correlated both with the decision to reopen (warmer states reopened earlier) and, plausibly, with transmission. It is therefore a variable one is entitled to condition on. Its practical bite here is limited, though, for a reason returned to below: almost no hot state stayed closed, so the untreated pool cannot span the warm-state temperature loading, and the event-time design does most of the work of separating a reopening effect from a coincident regional heat wave.

Mobility is the opposite. Reopening *causes* movement: lifting the order is precisely what permits people to drive, shop, and gather, so the post-reopening mobility surge is a downstream consequence of the treatment, a *mediator* on the causal path reopening $\rightarrow$ mobility $\rightarrow$ contact $\rightarrow$ transmission $\rightarrow$ cases. Conditioning on a mediator is post-treatment adjustment; it does not remove a confound, it blocks the channel through which the policy acts, biasing the estimated effect toward zero by construction (Rosenbaum, 1984; Cinelli et al., 2024). Conditioning on the treated state's own post-reopening mobility does collapse the estimated effect, and that collapse is the bad-control bias, not evidence of a null: reopening did its work *through* mobility, the mechanism Goolsbee and Syverson (2021) identify when they find that voluntary behavior, not the legal order itself, drove the pandemic's course. Mobility therefore enters this paper only as evidence on the *mechanism*, read off the temporal ordering (the mobility surge coincides with reopening and the case effect follows at the epidemiological lag), never as an adjustment.

The design principle that follows is simple. The shared pandemic factor is netted out through the untreated donor pool of the synthetic-interventions design; exogenous confounders such as temperature may in principle be conditioned on; post-treatment mediators such as mobility may not. One limitation of this program is structural rather than incidental: because almost no hot state stayed closed, the untreated donor pool cannot span the warm-state summer-temperature loading, so the thermal channel can be probed but never fully closed. That irreducible gap, the confounder one most needs to represent is the one the control group is least able to supply, is the honest signature of evaluating a policy that only one kind of state adopted, and it is the same limitation, in geographic rather than temporal form, that a coincident pandemic imposes on any single-shock design.

4.3.3 Synthetic interventions across the early reopeners

The synthetic-interventions (SI) estimator of Agarwal et al. (2026) extends synthetic control from a single control condition to *multiple* interventions. It rests on a tensor factor model in which a potential outcome under intervention d is $Y_{it}(d) = \langle \mathbf{u}_t(d), \mathbf{v}_i \rangle + \varepsilon_{it}(d)$, where the *unit* latent factors $\mathbf{v}_i$ are shared across time and across interventions, while the temporal factors $\mathbf{u}_t(d)$ carry the intervention. Identification of the target state's counterfactual under the stay-closed intervention rests on two conditions: selection on the latent factors, $\mathbb{E}[\varepsilon_{it}(d) \mid \mathbf{v}, \text{assignments}] = 0$, so that unconfoundedness holds given the shared factors rather than given observed covariates; and a linear-span condition, that the target state's loading $\mathbf{v}_i$ lies in the span of the loadings of the donor states that received the stay-closed intervention.

Operationally, for each early-reopening state as the target, SI regresses its pre-reopening trajectory onto the shelter-in-place donor pool, after denoising that pool's pre-treatment matrix by hard singular-value thresholding (a rank-k low-rank truncation), and applies the resulting weights to the donors' outcomes to impute the target's stay-closed counterfactual; the excess over that counterfactual is the reopening effect. Because the unit factors are shared across interventions, the target's loading is pinned on its own pre-reopening data and then read against the stay-closed temporal factors the donors trace. The default variant is the bias-corrected SI estimator, which reports an asymptotic-normality interval, though we lean on the cross-state and event-time pattern rather than that interval for inference, for reasons Section 4.5 makes explicit.

The design is deliberately plural. Rather than estimate the reopening effect for one state, we run SI with each of the eleven early-reopening states as the target in turn, against the same shelter-in-place donor pool, using each state's own reopening date. Eleven independently-timed reopenings, each imputed against states that stayed closed, convert a single case study into a replication: a genuine reopening effect should appear across the states, and, re-centered on each state's own reopening date, should switch on in event time rather than at a common calendar moment. That event-time alignment, and not a scalar average, is the object we report.

4.3.4 Reading the effect over time

In an epidemic a single average is a poor summary, and can be an actively misleading one. Reopening acts on confirmed cases only through a lagged pipeline, behavior to transmission to incubation to testing to a reported case, of roughly five to six weeks, so the effect is a delayed ramp rather than a level shift. The honest object is therefore the trajectory of the excess in event time: the cumulative excess as it accrues after each state's own reopening, and its daily rate, which reveals when the effect switches on. A flat pre-reopening excess is a placebo the design should pass; a post-reopening ramp that appears in *event* time, aligned across states with staggered reopening dates, is the signature of a policy effect rather than a coincident calendar-time wave.

4.4 Data

The primary outcome is the cumulative count of confirmed COVID-19 cases per 100{,}000 residents, constructed from the cumulative counts published by The New York Times (2021) and 2019 Census state populations. Cumulative incidence is inherently smooth (it integrates out weekday reporting noise) and yields an interpretable estimand, the excess cumulative cases per capita a reopening caused. We carry a second outcome through the identical design, cumulative deaths per 100{,}000 from the same source, so the reopening toll can be priced in lives and not only in infections; deaths follow infections at a lag of several weeks, so the end-of-window death effect is the leading edge of a toll that kept accruing past the window. The estimation window is 1 March through 30 June 2020.

The treated units are the eleven states that lifted statewide stay-at-home orders in late April or early May 2020 (Texas, Georgia, Florida, Arizona, South Carolina, Tennessee, Mississippi, Alabama, Oklahoma, Colorado, and Missouri), each entered on its own reopening date. The donor pool is the set of shelter-in-place-retaining states that did not reopen over the window, the untreated units whose trajectories carry the national epidemic factor without the reopening. Two auxiliary series enter the identification discussion of Section 4.3.2 but never the estimator's counterfactual: state-daily temperature (an exogenous confounder), from the county series of Bayat et al. (2020b) averaged to state; and Apple driving-mobility (a treatment mediator), which is read only for the mechanism.

4.5 Results

4.5.1 The reopening effect replicates across the early reopeners

Estimated state by state against the shelter-in-place donor pool, the reopening effect on cases is positive for all 11 of the eleven early reopeners, against small pre-treatment fits (RMSE 0.7 to 6.4 per 100k, set against effects in the hundreds). Table~4.1 reports each state's excess by 30 June. The median is 464 excess cases per 100k, ranging from the low hundreds to Arizona's 963, with Texas at 464. Converted to counts through each state's population and summed across the eleven, the early reopenings are associated with roughly 486,301 excess confirmed cases by the end of June. Run through the identical design, cumulative deaths tell the same story: the death effect is positive in all 11 states as well, and the eleven reopenings cost on the order of 10,009 excess deaths over the window, an independent outcome that both corroborates the case signal and prices the toll in lives. That eleven independently-timed reopenings, each imputed from states that stayed closed, all point the same way on both outcomes is the substance of the result: it is a replication, not a single case study.

The design's identifying power, though, is in the timing, not the magnitude. Figure 4.1 re-centers each state's excess on its own reopening date. Aligned in event time (left), the

State	Reopened	Cases / 100k	Excess cases	Deaths / 100k	Excess deaths
Arizona	08 May	963	70,132	16.8	1,224
Alabama	30 Apr	612	29,973	13.1	644
Mississippi	27 Apr	547	16,303	26.1	779
South Carolina	20 Apr	535	27,541	10.0	514
Florida	04 May	510	109,624	9.0	1,945
Texas	01 May	464	134,553	5.3	1,534
Tennessee	27 Apr	428	29,217	4.7	321
Georgia	24 Apr	332	35,184	13.8	1,461
Colorado	27 Apr	220	12,685	14.6	841
Missouri	04 May	210	12,888	10.2	625
Oklahoma	24 Apr	207	8,202	3.1	122
Total (11 states)			486,301		10,009

Table 4.1: Synthetic-interventions estimate of the reopening effect by 30 June 2020: excess cumulative COVID-19 cases and deaths, per 100,000 and as raw counts, for each early-reopening state against the shelter-in-place donor pool. Pre-treatment fits are reported in the text.

eleven trajectories share one shape: a pre-reopening excess indistinguishable from zero (mean -0.2 per 100k over the four weeks before each state's reopening, the placebo the design should pass), then a ramp that builds to a mean of 255 excess cases per 100k some seven to eight weeks out. The flat pre-period is decisive against the obvious objection, that this is merely the Sun Belt's summer wave: even the states that reopened latest show no excess before their own reopening date, which a common calendar-time heat wave could not produce. Aligned in calendar time (right), the same trajectories are offset by their staggered reopening dates, and the shared event-time shape dissolves, exactly the pattern a reopening effect, and not a regional coincidence, implies.

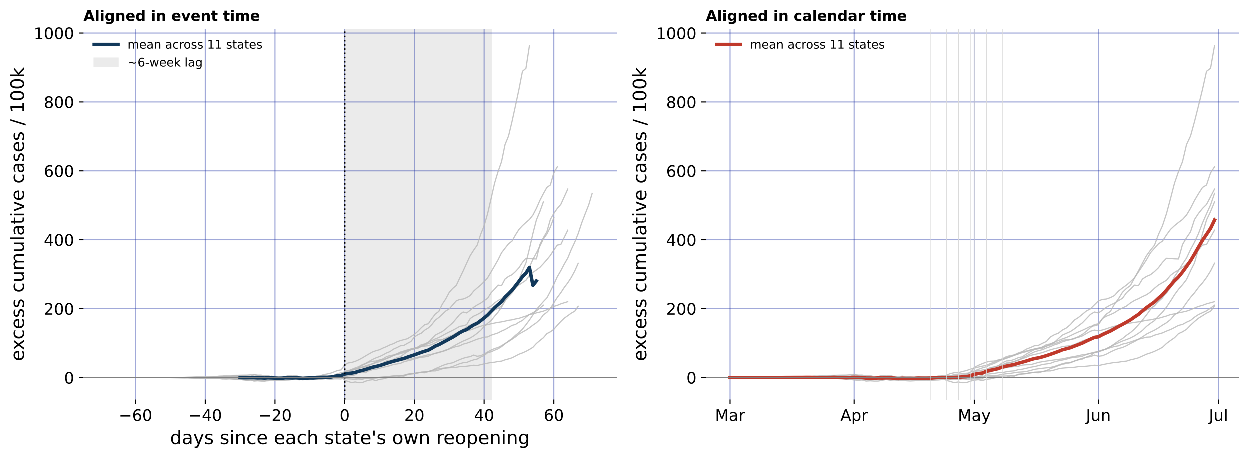


Figure 4.1: Reopening effect across the 11 early-reopening states (synthetic interventions): excess cumulative cases per 100k, re-centered on each state's own reopening date (left, event time) and in calendar time (right). Grey: individual states. Bold: mean. The pre-reopening excess is flat at zero; the ramp appears in event time but not calendar time.

4.5.2 The mechanism: reopening acts through mobility

Mobility is a mediator, not a control, so it enters not as an adjustment but as evidence on *how* reopening moves cases (Section 4.3.2). Figure 4.2 overlays, in event time, the driving-mobility path across the early reopeners with the daily rate of the case effect. Mobility jumps at the reopening date, from a locked-down mean of 70 in the three weeks before reopening to 98 in the three weeks after (relative to a January baseline of 100), and the case-effect rate lifts off the axis several weeks later, at the incubation-and-testing lag. The temporal ordering, movement first, cases after, is the signature of the pathway reopening → mobility → transmission → cases, and it is why conditioning on the post-reopening mobility surge would net the effect away: it is the channel, not a confound.

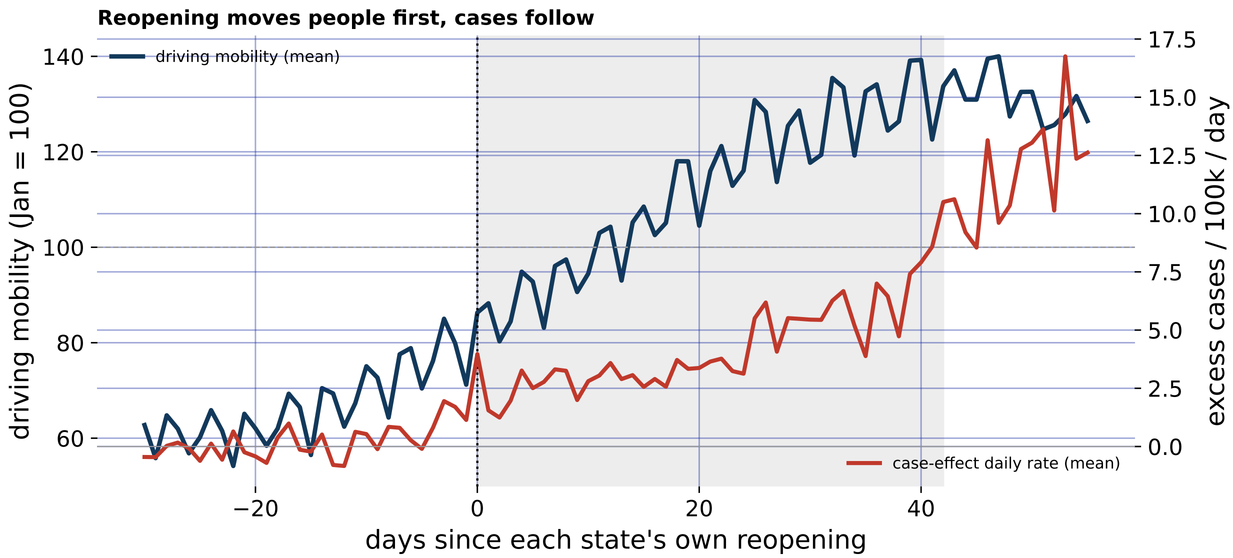


Figure 4.2: Mechanism, in event time across the 11 early reopeners: driving mobility (left axis) surges at reopening, and the daily rate of the case effect (right axis) follows at the epidemiological lag.

4.6 Conclusion

Reopening in the spring of 2020 was a policy applied in the middle of a coincident, non-stationary pandemic, and separating its effect from the epidemic's own trajectory is the identification problem this chapter shares with a border closure: the treated unit's own history cannot supply the counterfactual, so the comparison must run through states that felt the same shock but did not reopen. Synthetic interventions supply exactly that comparison, imputing each early reopener's stay-closed counterfactual from the shelter-in-place pool, and estimated across eleven independently-timed reopenings the effect is uniformly positive: a median of roughly 464 excess cumulative cases per 100{,}000 by the end of June, and larger in the hardest-hit states. Priced in counts and summed across the eleven, that is on the order of 486,301 excess confirmed cases, and, on the identical design run on deaths, some 10,009 excess deaths, by 30 June alone. Both totals are conservative: confirmed cases understate true infections under the thin testing of spring 2020, and because deaths lag infections by weeks, the end-of-June death effect is only the leading edge of a toll that kept accruing into the summer.

The design's strongest evidence is in event time. The excess is flat before each state's own reopening and ramps afterward, and because the reopening dates are staggered, that ramp appears in event time but not in calendar time, which is what a genuine policy effect, and not the Sun Belt's coincident summer wave, implies. The pathway is legible in the same alignment: driving mobility surges at reopening and the case effect follows at the epidemiological lag, so reopening acts on cases through the behavior it unleashes, exactly as Goolsbee and Syverson (2021) argued and as the mediator logic requires. Two honest limits remain, and both are the signature of evaluating a policy that only one kind of state adopted: with almost no hot state left closed, the thermal channel can be probed but not fully closed, and because a mediator cannot be conditioned on, the effect is estimated through, not net of, the mobility it induces.

Because synthetic interventions impute a unit's outcome under an intervention it did not receive, the mirror question is in principle also askable: how the shelter-in-place states would have looked had they reopened, borrowing the reopen factors from the states that did. That reverse imputation is not identified here, and the estimator says so plainly. The linear-span condition that lets a Sun Belt reopener be written in terms of the large, diverse stay-closed pool does not hold turned around: the eleven reopeners are too narrow a set to span the early-epidemic control states, whose March explosion no reopener resembles. The pre-treatment fit that is flat and small in the forward direction (RMSE at most 6 per 100{,}000) breaks down in reverse, reaching 56 per 100{,}000 for New York and failing the pre-reopening placebo for precisely those hard-hit states, so the reverse effect is positive for only 7 of the fourteen controls and cannot be read as a symmetric confirmation. This asymmetry is not a defect of the method but its identification condition made visible: synthetic interventions recover a counterfactual only in the direction where the donor pool spans the target, and the pre-treatment RMSE is the diagnostic that flags the other direction as off-support. That the reopening question is answerable forward but not backward is the same one-kind-of-state limitation, in its sharpest form, that runs through the chapter. Within those limits, the reading is clear, and only visible in hindsight: the early reopenings were followed, at the epidemiological lag, by a large and replicable excess of COVID-19 cases, and of deaths, that the states that stayed closed did not incur.

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